The critical O(N) CFT: Methods and conformal data. Updated tables

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Abstract: This note contains up-to-date tables of [1]. Superseded results are displayed in gray, new results in orange.

Please email the author to report any omissions.

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Contents

1 Summary

We give references to papers that appeared after 24 January 2022 (and papers which appeared before that date but which were accidentally omitted).

1.1 New perturbative studies

- We have made a few computations specifically for the preparation of this note and which are not presented elsewhere. They are cited as [2].
- Many high-loop results for operators in the ε-expansion were computed in [3][1] and [4, 5].[2]

[1]Appeared before January 2022 but they were not include it in [1] due to oversight.

[2]Complete results of [5] are released in [6], however we only give reference to [5].

- The OPE coefficient involving `Op[E,2,2]` and two ϕ was determined exactly in [7].
- The seven-loop results, including $O(\varepsilon^8)$ for Δ_ϕ, have been presented in [8].
- [9]: New result for the $O(N)$ free energy.

1.2 New MC studies

- Conventional MC studies: Ising [10], $O(2)$ [11] $O(3)$ [12].
- Fuzzy sphere studies: Ising [13–17], $O(2)$ [18][3], $O(3)$ [19, 20].

1.3 New bootstrap studies

- Ising [21–24].

1.4 New formal studies

- Status of the $2+\tilde{\epsilon}$ expansion [25].

2 New implementations

We have implemented many new operators in the data file `ONdata.m`. This now contains all Ising scalar operators with $\Delta_{4d} \leqslant 12$, all Ising operators with $\Delta_{4d} \leqslant 10$, all $O(N)$ scalar operators with $\Delta_{4d} \leqslant 8$, all $O(N)$ spin-ℓ operators with $\Delta_{4d} \leqslant 7$, and remaining (Lorentz non-traceless-symmetric) $O(N)$ operators with $\Delta_{4d} \leqslant 6$. In addition several operators with higher scaling dimension has been included in a non-systematic fashion. On the tables below, operator names are displayed in orange where new results are added.

We have also implemented the critical coupling as

$$\texttt{ValueE[criticalCoupling]} = \frac{3\texttt{e}}{\texttt{n}+8} + \frac{9(3\texttt{n}+14)\texttt{e}^2}{(\texttt{n}+8)^3} + \dots$$

[3] https://scgp.stonybrook.edu/video_portal/results.php?event_id=437.

3 Tables

Table 1: Estimates and measurements of the critical exponents η and ν in the three-dimensional Ising CFT.

		η	ν
Free theory		0	0.5
Wilson–Fisher $O(\varepsilon^3)$ truncation		0.037209	0.60750
Wilson–Fisher $O(\varepsilon^7)$ resummation	[26]	0.03653(65)	0.62977(22)
Uniaxial magnet FeF_2 (1987)	[27]		0.64(1)
Liquid-gas transition CO_2 (1998)	[28]	0.042(6)	
Liquid-gas transition D_2O (2000)	[29]		0.62(3)
Liquid mixture* (1988)	[30]	0.032(13)	0.629(3)
Liquid mixture† (2004)	[31]	0.041(5)	0.632(2)
Quantum phase transition (2020)	[32]		0.62(4)
High temperature expansion	[33]	0.03639(15)	0.63012(16)
MC simulations	[34]	0.036284(40)	0.62998(5)
Numerical bootstrap	[35]	0.0362978(20)	0.629971(4)
Numerical bootstrap	[23]	0.036297612(48)	0.62997097(12)

*Light-scattering measurement in a mixture of water and isobutyric acid.
†Turbidity measurement in a mixture of methanol and cyclohexane.

Table 3: A summary of the developments of higher order results for the main observables in the ε-expansion and $1/N$ expansion. The notation is "order[ref.]".

$d = 4 - \varepsilon$ expansion		$1/N$ expansion	
Δ_φ	ε^4[36]ε^5[37, 38]$^*\varepsilon^6$[39]ε^7[40]ε^8[41]	Δ_φ	N^{-2}(3d)[42]N^{-2}[43, 44]N^{-3}[45]†
$\Delta_{\varphi^2_S}$	ε^4[46]ε^5[38]ε^6[47]ε^7[40]	Δ_σ	N^{-2}(3d)[48]N^{-2}[49]
$\Delta_{\varphi^4_S}$	ε^4[46]ε^5[38]ε^6[47]ε^7[40]	Δ_{σ^2}	N^{-1}[50]$^\ddagger N^{-2}$[51, 52]
$\Delta_{\varphi^2_T}$	ε^2[53]ε^3[54]ε^4[55]ε^6[56]§	$\Delta_{\varphi^2_T}$	N^{-1}[50]N^{-2}[57]
$\Delta_{\varphi^4_{T_4}}$	ε^5[58, 59]ε^6[60]¶	$\Delta_{\varphi^4_{T_4}}$	N^{-1}[61]N^{-2}[62]
C_T	ε^2[63]$^\Vert\varepsilon^3$[64]ε^4[65]	C_T	N^{-1}[66]
C_J	ε^2[67]$^{**}\varepsilon^3$[64]ε^4[65]	C_J	N^{-1}(3d)[68]N^{-1}[69]

*The result in [37] contained several errors, see chapter 15 of [70].

†As noted in [71], the expression in [45] contains a misprint in equation (22). See also [72, 73].

‡Ref. [74] (page 221) cites [50] together with Bervillier, Girardi, Brezin; Saclay preprint (1974).

§Note that their $\zeta_{3,5}$ equals our $\zeta_{5,3}$.

¶This result was extracted from the more general results of [60].

$^{\Vert,**}$Can be extracted from [75], see comment in [67].

Table 5: Results in three dimensions from the conformal bootstrap. The lower part of the table contains the critical exponents, computed from the values in the upper part. For comparison, we have added the exact results at $N=\infty$ (sometimes denoted the spherical model).

Quantity	$N=1$		$N=2$		$N=3$		$N=\infty$
Δ_φ (Δ_σ)	0.518148806(24)	[23]	0.519088(22)	[76]	0.518942(51)	[77]	0.5
$\Delta_{\varphi_S^2}$ (Δ_ϵ)	1.41262528(29)	[23]	1.51136(22)	[76]	1.59489(59)	[77]	2
$\Delta_{\varphi_S^4}$ $(\Delta_{\epsilon'})$	3.82951(61)	[78]	3.794(8)	[79]	3.7668(100)	[80]*	4
$\Delta_{\varphi_T^2}$ (Δ_t)			1.23629(11)	[76]	1.20954(32)	[77]	1
$\Delta_{\varphi_{T_4}^4}$ (Δ_{t_4})			3.11535(73)	[76]	$<2.99056^\dagger$	[77]	2
$\dfrac{C_T}{C_{T,\text{free}}}$	0.946543(42) 0.946538675(42)	[78] [23]	0.944056(15)	[76]	0.944524(28)	[77]	1
$\dfrac{C_J}{C_{J,\text{free}}}$			0.904395(28)	[76]	0.90632(16)	[77]	1
α	0.11008708(35)		$-0.01526(30)$		$-0.1350(12)$		-1
β	0.32641871(6)		0.348699(54)		0.36932(22)		0.5
γ	1.23707551(23)		1.31786(20)		1.39641(81)		2
δ	4.78984254(23)		4.77937(24)		4.78106(75)		5
η	0.036297612(48)		0.038176(44)		0.03787(13)		0
ν	0.62997097(12)	[23]	0.671754(99)	[76]	0.71168(41)	[77]	1
ϕ_c			1.18478(19)	[76]	1.27424(83)	[77]	2
ω	0.82951(61)	[78]	0.794(8)	[79]	0.7668(100)	[80]	1

*Computed using $\Lambda = 35$ data from 42 sampling points [80].
†This value is a rigorous upper bound and the true operator dimension is expected to lie not far from this value.

Table 6: Results in three dimensions from Monte Carlo simulations. For $N = 0$ the exponent ν and the Hausdorff dimension $D_H = d - \Delta_{\varphi_T^2}$ are directly related.

Quantity	$N = 0$		$N = 1$		$N = 2$		$N = 3$	
γ	1.15695300(95)	[81]						
η	0.0250(14)	[82]	0.036284(40)	[34]	0.03810(8)	[83]	0.03784(5)	[84]
ν	0.58759700(40)	[85]	0.62998(5)	[34]	0.67169(7)	[83]	0.71164(10)	[84]
ω	0.9037(56)	[86]	0.832(6)	[87]	0.789(4)	[83]	0.759(2)	[84]
d_H	1.7018467(16)*	[85]	1.7349(65)	[88]	1.7626(66)	[88]	1.7906(3)	[89]

*Computed from $\Delta_{\varphi_T^2}$.

3.1 Families of operators for $N = 1$

`IsingF1[`k`]` Scalar operators of the form ϕ^k, $k \geqslant 1$, $k \neq 3$. We have ε^2[90] ε^5[5, 91],[4]

$$\texttt{DeltaE[IsingF1[}k\texttt{]]} = k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{k(k-1)}{6}\varepsilon - \frac{k(17k^2 - 67k + 47)}{324}\varepsilon^2 + \ldots + O(\varepsilon^6). \quad (3.1)$$

`IsingF2[`l`]` Broken currents of the form $\mathcal{J}_\ell$, $\ell \in 2\mathbb{Z}_+$. We have ε^2[92]ε^4[93],

$$\texttt{DeltaE[IsingF2[}l\texttt{]]} = 2 + l - \varepsilon + \left(\frac{1}{54} - \frac{1}{9l(l+1)}\right)\varepsilon^2 + \ldots + O(\varepsilon^5). \quad (3.2)$$

`IsingF3[`l`]` Spinning operators of the form $\partial^\ell \phi^3$, $\ell \geqslant 2$, ε^1[94]ε^2[7],

$$\texttt{DeltaE[IsingF3[}l\texttt{]]} = 3 + l - \frac{3\varepsilon}{2} + \frac{\varepsilon}{3}\left(1 + \frac{2(-1)^l}{l+1}\right) + \ldots + O(\varepsilon^3). \quad (3.3)$$

Starting from $\ell = 6$, there are additional operators of the type $\partial^\ell \phi^3$, however they all have vanishing first-order anomalous dimensions. The family `IsingF3` is identified by the non-vanishing of the anomalous dimension to order ε.

`IsingF4[`k`]` Spin-two operators of the form $\partial^2 \phi^k$, $k \geqslant 2$, ε^1[95]ε^2[6, 91],[5]

$$\texttt{DeltaE[IsingF4[}k\texttt{]]} = 2 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{(k-2)(3k+1)}{18}\varepsilon - \frac{(k-2)(102k^2 - 335k - 235)}{1944}\varepsilon^2 + O(\varepsilon^3). \quad (3.4)$$

`IsingF5[`k`]` Spin-three operators of the form $\partial^3 \phi^k$, $k \geqslant 3$, $k \neq 4$, ε^1[95],

$$\texttt{DeltaE[IsingF5[}k\texttt{]]} = 3 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{k^2 - 2k - 2}{6}\varepsilon + O(\varepsilon^2). \quad (3.5)$$

`IsingF6[`k`]` Spin-four operators of the form $\partial^4 \phi^k$, $k \geqslant 2$, ε^1[95],

$$\texttt{DeltaE[IsingF6[}k\texttt{]]} = 4 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{(k-2)(5k-1)}{30}\varepsilon + O(\varepsilon^2). \quad (3.6)$$

`IsingF7[`k`]` Spin-four operators of the form $\partial^4 \phi^k$, $k \geqslant 4$, ε^1[95],

$$\texttt{DeltaE[IsingF7[}k\texttt{]]} = 4 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{3k^2 - 7k - 12}{18}\varepsilon + O(\varepsilon^2). \quad (3.7)$$

`IsingF8[`k`]` Spin-five operators of the form $\partial^5 \phi^k$, $k \geqslant 3$, $k \neq 4$, ε^1[95],

$$\texttt{DeltaE[IsingF8[}k\texttt{]]} = 5 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{3k^2 - 7k - 2}{18}\varepsilon + O(\varepsilon^2). \quad (3.8)$$

[4] The formula was first found using results in [5], then independently in [91].

[5] The formula was derived in [6] by combining results from [5] and [91]

`IsingF9[`k`]` Spin-five operators of the form $\partial^5\phi^k$, $k \geqslant 5$, ε^1[95],

$$\texttt{DeltaE[IsingF9[}k\texttt{]]} = 5 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{3k^2 - 8k - 20}{18}\varepsilon + O(\varepsilon^2). \tag{3.9}$$

`IsingF10[`k`]` Scalar operators of the form $\Box^2\phi^k$, $k \geqslant 4$, ε^1[95]ε^2[6, 91],[6],

$$\texttt{DeltaE[IsingF10[}k\texttt{]]} = 4 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{k(3k-7)}{18}\varepsilon - \frac{51k^3 - 338k^2 + 443k + 108}{972} + O(\varepsilon^3). \tag{3.10}$$

`IsingF11[`k`]` Spin-one operators of the form $\partial\Box^2\phi^k$, $k \geqslant 5$, ε^1[95],

$$\texttt{DeltaE[IsingF11[}k\texttt{]]} = 5 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{3k^2 - 8k - 4}{18}\varepsilon + O(\varepsilon^2) \tag{3.11}$$

`IsingF12[`k`]` Spin-two operators of the form $\partial^2\Box\phi^k$, $k \geqslant 4$, ε^1[95],

$$\texttt{DeltaE[IsingF12[}k\texttt{]]} = 4 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{3k^2 - 7k - 4}{18}\varepsilon + O(\varepsilon^2). \tag{3.12}$$

`IsingF13[`k`]` Spin-three operators of the form $\partial^3\Box\phi^k$, $k \geqslant 5$, ε^1[95],

$$\texttt{DeltaE[IsingF13[}k\texttt{]]} = 5 + k\left[1 - \tfrac{\varepsilon}{2}\right] + \frac{3k^2 - 8k - 10}{18}\varepsilon + O(\varepsilon^2). \tag{3.13}$$

[6] The formula was derived in [6] by combining results from [5] and [91]

Table 8: $\mathbb{Z}_2$ even scalar operators for $N = 1$. The table includes operators with $\Delta^{4d} \leqslant 12$. The fact that there are two operators with anomalous dimension $\frac{8}{3}$ of the form $\Box^3\phi^6$ is not a typo, for instance both operators are reported in table 5 of [95].

$\mathcal{O}$	$\mathcal{O}\lvert\varepsilon$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	Δ_{3d}	Family
`Op[E,0,0]`	$\mathbb{1}$	0	exact	0	
`Op[E,0,1]`	ϕ^2	$[2-\varepsilon]+\frac{1}{3}\varepsilon$	ε^7[40]	1.412625(10)[35]	1_2
`Op[E,0,2]`	ϕ^4	$[4-2\varepsilon]+2\varepsilon$	ε^7[40]	3.82951(61)[78]	1_4
`Op[E,0,3]`	ϕ^6	$[6-3\varepsilon]+5\varepsilon$	ε^2[90]ε^5[3]	6.8956(43)[96]	1_6
`Op[E,0,4]`	$\Box^2\phi^4$	$[8-2\varepsilon]+\frac{10}{9}\varepsilon$	ε^4[5]	7.2535(51)[96]	10_4
`Op[E,0,5]`	ϕ^8	$[8-4\varepsilon]+\frac{28}{3}\varepsilon$	ε^2[90]ε^5[5]		1_8
`Op[E,0,6]`	$\Box^3\phi^4$	$[10-2\varepsilon]+\frac{1}{3}\varepsilon$	ε^2[4]		
`Op[E,0,7]`	$\Box^2\phi^6$	$[10-3\varepsilon]+\frac{11}{3}\varepsilon$	$\varepsilon^1\varepsilon^2$[4]		10_6
`Op[E,0,8]`	ϕ^{10}	$[10-5\varepsilon]+15\varepsilon$	ε^2[90]ε^5[5]		1_{10}
`Op[E,0,9]`	$\Box^4\phi^4$	$[12-2\varepsilon]+\frac{14}{15}\varepsilon$	ε^1		
`Op[E,0,10]`	$\Box^3\phi^6$	$[12-3\varepsilon]+\frac{8}{3}\varepsilon$	ε^1		
`Op[E,0,11]`	$\Box^3\phi^6$	$[12-3\varepsilon]+\frac{8}{3}\varepsilon$	ε^1		
`Op[E,0,12]`	$\Box^2\phi^8$	$[12-4\varepsilon]+\frac{68}{9}\varepsilon$	ε^2[6, 91]		10_8
`Op[E,0,13]`	ϕ^{12}	$[12-6\varepsilon]+22\varepsilon$	ε^2[90]ε^5[5]		1_{12}

Table 9: $\mathbb{Z}_2$ odd scalar operators for $N = 1$. The table includes operators with $\Delta^{4d} \leqslant 11$.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	Δ_{3d}	Family
`Op[0,0,1]`	ϕ	$[1 - \frac{1}{2}\varepsilon] + 0\varepsilon$	ε^8[41]	0.5181489(10)[35]	1_1
`Op[0,0,2]`	ϕ^5	$[5 - \frac{5}{2}\varepsilon] + \frac{10}{3}\varepsilon$	ε^3[97]ε^5[5]	5.2906(11)[96]*	1_5
`Op[0,0,3]`	ϕ^7	$[7 - \frac{7}{2}\varepsilon] + 7\varepsilon$	ε^2[90]ε^5[5]		1_7
`Op[0,0,4]`	$\Box^2\phi^5$	$[9 - \frac{5}{2}\varepsilon] + \frac{20}{9}\varepsilon$	ε^3[5]		10_5
`Op[0,0,5]`	ϕ^9	$[9 - \frac{9}{5}\varepsilon] + 12\varepsilon$	ε^2[90]ε^5[5]		1_9
`Op[0,0,6]`	$\Box^3\phi^5$	$[11 - \frac{5}{2}\varepsilon] + \frac{4}{3}\varepsilon$	ε^1		
`Op[0,0,7]`	$\Box^2\phi^7$	$[11 - \frac{7}{2}\varepsilon] + \frac{49}{9}$	ε^2[6, 91]		10_7
`Op[0,0,8]`	ϕ^{11}	$[11 - \frac{11}{2}\varepsilon] + \frac{55}{3}\varepsilon$	ε^2[90]ε^5[5]		1_{11}

*A value with rigorous error bars, 5.262(89), was given in [78].

Table 10: $\mathbb{Z}_2$ even operators of spin two, for $N = 1$. The table includes operators with $\Delta_{4d} \leqslant 10$.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	Δ_{3d}	Family
`Op[E,2,1]`	$T^{\mu\nu}$	$[4-\varepsilon]$	exact	3	2_2, 4_2
`Op[E,2,2]`	$\partial^2\phi^4$	$[6-2\varepsilon]+\frac{13}{9}\varepsilon$	ε^5[5]	5.50915(44)[96]*	4_4
`Op[E,2,3]`	$\partial^2\Box\phi^4$	$[8-2\varepsilon]+\frac{8}{9}\varepsilon$	ε^1	7.0758(58)[96]	12_4
`Op[E,2,4]`	$\partial^2\phi^6$	$[8-3\varepsilon]+\frac{38}{9}\varepsilon$	ε^2[6, 91]		4_6
`Op[E,2,5]`	$\partial^2\Box^2\phi^4$	$[10-2\varepsilon]+\frac{1}{9}\varepsilon$	ε^1		
`Op[E,2,6]`	$\partial^2\Box^2\phi^4$	$[10-2\varepsilon]+\frac{14}{15}\varepsilon$	ε^1		
`Op[E,2,7]`	$\partial^2\Box\phi^6$	$[10-3\varepsilon]+\frac{31}{9}\varepsilon$	ε^1		12_6
`Op[E,2,8]`	$\partial^2\phi^8$	$[10-4\varepsilon]+\frac{25}{3}\varepsilon$	ε^2[6, 91]		4_8

*A value with rigorous error bars, 5.499(17), was given in [78].

Table 11: $\mathbb{Z}_2$ even operators of spin four, for $N = 1$. The table includes operators with $\Delta^{4\text{d}} \leqslant 10$.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta_{3\text{d}}$	Family
`Op[E,4,1]`	$C^{\mu\nu\rho\sigma}$	$[6-\varepsilon] + 0\varepsilon$	ε^4[93]	5.022665(28)[96]	2_4, 6_2
`Op[E,4,2]`	$\partial^4\phi^4$	$[8-2\varepsilon] + \frac{4}{9}\varepsilon$	ε^1	6.42065(64)[96]	7_4
`Op[E,4,3]`	$\partial^4\phi^4$	$[8-2\varepsilon] + \frac{19}{15}\varepsilon$	ε^1	7.38568(28)[96]	6_4
`Op[E,4,4]`	$\partial^4\Box\phi^4$	$[10-2\varepsilon] + \frac{49-\sqrt{697}}{90}\varepsilon$	ε^1	8.9410(99)[96]	
`Op[E,4,5]`	$\partial^4\Box\phi^4$	$[10-2\varepsilon] + \frac{49+\sqrt{697}}{90}\varepsilon$	ε^1		
`Op[E,4,6]`	$\partial^4\phi^6$	$[10-3\varepsilon] + \frac{58}{15}\varepsilon$	ε^1		6_6
`Op[E,4,7]`	$\partial^4\phi^6$	$[10-3\varepsilon] + 3\varepsilon$	ε^1		7_6

Table 12: $\mathbb{Z}_2$ even operators for $N = 1$ of leading twist for even spin. The identification with numerical data can only be done for the smallest operator at each spin. For spin 6 and 8 we also include operators of subleading twist $\tau^{4\mathrm{d}} = 4$.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta_{3\mathrm{d}}$	Family
`Op[E,6,1]`	$\mathcal{J}_6$	$[8-\varepsilon]+0\varepsilon$	ε^4[93]	7.028488(16)[96]	2_7
`Op[E,6,2]`	$\partial^6\phi^4$	$[10-2\varepsilon]+0.4966\varepsilon$	ε^1		
`Op[E,6,3]`	$\partial^6\phi^4$	$[10-2\varepsilon]+0.6707\varepsilon$	ε^1		
`Op[E,6,4]`	$\partial^6\phi^4$	$[10-2\varepsilon]+1.1787\varepsilon$	ε^1		
`Op[E,8,1]`	$\mathcal{J}_8$	$[10-\varepsilon]+0\varepsilon$	ε^4[93]	9.031023(30)[96]	2_8
`Op[E,8,2]`	$\partial^8\phi^4$	$[12-2\varepsilon]+0.1142\varepsilon$	ε^1		
`Op[E,8,3]`	$\partial^8\phi^4$	$[12-2\varepsilon]+0.5217\varepsilon$	ε^1		
`Op[E,8,4]`	$\partial^8\phi^4$	$[12-2\varepsilon]+0.6752\varepsilon$	ε^1		
`Op[E,8,5]`	$\partial^8\phi^4$	$[12-2\varepsilon]+1.1276\varepsilon$	ε^1		
`Op[E,10,1]`	$\mathcal{J}_{10}$	$[12-\varepsilon]+0\varepsilon$	ε^4[93]	11.0324141(99)[96]	2_{10}
`Op[E,12,1]`	$\mathcal{J}_{12}$	$[14-\varepsilon]+0\varepsilon$	ε^4[93]	13.033286(12)[96]	2_{12}
`Op[E,14,1]`	$\mathcal{J}_{14}$	$[16-\varepsilon]+0\varepsilon$	ε^4[93]	15.033838(15)[96]	2_{14}
`Op[E,16,1]`	$\mathcal{J}_{16}$	$[18-\varepsilon]+0\varepsilon$	ε^4[93]	17.034258(34)[96]	2_{16}

Table 13: $\mathbb{Z}_2$ even operators for $N = 1$ of leading twist for odd spin.

$\mathcal{O}$	$\mathcal{O}\|\varepsilon$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta_{3\mathrm{d}}$	Family
`Op[E,1,1]`	$\partial\Box^2\phi^6$	$[11-3\varepsilon]+\frac{28}{9}\varepsilon$	ε^1		$\mathtt{11}_6$
`Op[E,3,1]`	$\partial^3\phi^6$	$[9-3\varepsilon]+\frac{11}{3}\varepsilon$	ε^1		$\mathtt{5}_6$
`Op[E,5,1]`	$\partial^5\Box\phi^4$	$[11-2\varepsilon]+\frac{29}{45}\varepsilon$	ε^1		
`Op[E,5,2]`	$\partial^5\phi^6$	$[11-3\varepsilon]+\frac{20}{9}\varepsilon$	ε^1		$\mathtt{9}_6$
`Op[E,5,3]`	$\partial^5\phi^6$	$[11-3\varepsilon]+\frac{32}{9}\varepsilon$	ε^1		$\mathtt{8}_6$
`Op[E,7,1]`	$\partial^7\phi^4$	$[11-2\varepsilon]+\frac{1}{3}\varepsilon$	ε^1		
`Op[E,9,1]`	$\partial^9\phi^4$	$[13-2\varepsilon]+\frac{1}{3}\varepsilon$	ε^1		
`Op[E,9,2]`	$\partial^9\phi^4$	$[13-2\varepsilon]+\frac{5}{9}\varepsilon$	ε^1		

Table 14: A selection of spinning $\mathbb{Z}_2$ odd operators for $N=1$. The table includes all such operators with $\Delta^{4\mathrm{d}} \leqslant 10$.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta_{3\mathrm{d}}$	Family
`Op[0,1,1]`	$\partial\Box^2\phi^5$	$[10-\frac{5}{2}\varepsilon]+\frac{31}{18}\varepsilon$	ε^1		11_5
`Op[0,2,1]`	$\partial^2\phi^3$	$[5-\frac{3}{2}\varepsilon]+\frac{5}{9}\varepsilon$	ε^2[7]ε^5[5]	4.180305(18)[96]	3_2, 4_3
`Op[0,2,2]`	$\partial^2\phi^5$	$[7-\frac{5}{2}\varepsilon]+\frac{8}{3}\varepsilon$	ε^2[6, 91]	6.9873(53)[96]	4_5
`Op[0,2,3]`	$\partial^2\Box\phi^5$	$[9-\frac{5}{2}\varepsilon]+2\varepsilon$	ε^1		12_5
`Op[0,2,4]`	$\partial^2\phi^7$	$[9-\frac{7}{2}\varepsilon]+\frac{55}{9}\varepsilon$	ε^2[6, 91]		4_7
`Op[0,3,1]`	$\partial^3\phi^3$	$[6-\frac{3}{2}\varepsilon]+\frac{1}{6}\varepsilon$	ε^2[7]	4.63804(88)[96]	3_3, 5_3
`Op[0,3,2]`	$\partial^3\phi^5$	$[8-\frac{5}{2}\varepsilon]+\frac{13}{6}\varepsilon$	ε^1		5_5
`Op[0,3,3]`	$\partial^3\Box\phi^5$	$[10-\frac{5}{2}\varepsilon]+\frac{25}{18}\varepsilon$	ε^1		13_5
`Op[0,3,4]`	$\partial^3\phi^7$	$[10-\frac{7}{2}\varepsilon]+\frac{11}{2}\varepsilon$	ε^1		5_7
`Op[0,4,1]`	$\partial^4\phi^3$	$[7-\frac{3}{2}\varepsilon]+\frac{7}{15}\varepsilon$	ε^2[7]	6.112674(19)[96]	3_4, 6_3
`Op[0,4,2]`	$\partial^4\phi^5$	$[9-\frac{5}{2}\varepsilon]+\frac{14}{9}\varepsilon$	ε^1		7_5
`Op[0,4,3]`	$\partial^4\phi^5$	$[9-\frac{5}{2}\varepsilon]+\frac{12}{5}\varepsilon$	ε^1		6_5
`Op[0,5,1]`	$\partial^5\phi^3$	$[8-\frac{3}{2}\varepsilon]+\frac{2}{9}\varepsilon$	ε^2[7]	6.709778(27)[96]	3_5, 8_3
`Op[0,5,2]`	$\partial^5\phi^5$	$[10-\frac{5}{2}\varepsilon]+\frac{5}{6}\varepsilon$	ε^1		9_5
`Op[0,5,3]`	$\partial^5\phi^5$	$[10-\frac{5}{2}\varepsilon]+\frac{19}{9}\varepsilon$	ε^1		8_5
`Op[0,6,1]`	$\partial^6\phi^3$	$[9-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^2[2, 98]		
`Op[0,6,2]`	$\partial^6\phi^3$	$[9-\frac{3}{2}\varepsilon]+\frac{3}{7}\varepsilon$	ε^2[7]		3_6
`Op[0,7,1]`	$\partial^7\phi^3$	$[10-\frac{3}{2}\varepsilon]+\frac{1}{4}\varepsilon$	ε^2[7]	8.747293(56)[96]	3_7
`Op[0,8,1]`	$\partial^8\phi^3$	$[11-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^2[2, 98]		
`Op[0,8,2]`	$\partial^8\phi^3$	$[11-\frac{3}{2}\varepsilon]+\frac{11}{27}\varepsilon$	ε^2[7]		3_8
`Op[0,9,1]`	$\partial^9\phi^3$	$[12-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^2[2, 98]		
`Op[0,9,2]`	$\partial^9\phi^3$	$[12-\frac{3}{2}\varepsilon]+\frac{4}{15}\varepsilon$	ε^2[7]		3_9

New two-loop results for twist-3 operators can be computed by a two-loop dilatation operator, whose form was guessed based on analogy with [99], and confirmed in [98]. For these results we write [2, 98].

Note that this produces negative anomalous dimensions; the first instance is `Op[0,6,1]` with

$$\texttt{DeltaE[Op[0,6,1]]} = \left[9-\frac{3\texttt{e}}{2}\right]-\frac{\texttt{e}^2}{324}+\texttt{ord e}^2 \tag{3.14}$$

The data file has implemented these operators up to spin 13.

3.2 Operators in non-traceless-symmetric Lorentz representations

Table 15: Lorentz non-traceless-symmetric operators with $\Delta^{4\text{d}} \leqslant 10$. The 4d notation for the representations is the one used in [95]. In the data file, these operators have now been implemented, e.g. `DeltaE[Op[0,{2,2},1]]`

Lorentz YT	4d representation	Field content	$\Delta^{(1)}_{4-\varepsilon}$
$\ydiagram{2,2}$	$(2,0)+(0,2)$	$\partial^4\phi^3$	$[7-\frac{3}{2}\varepsilon]+0\varepsilon$
$\ydiagram{4,1}$	$(\frac{5}{2},\frac{3}{2})+(\frac{3}{2},\frac{5}{2})$	$\partial^5\phi^3$	$[8-\frac{3}{2}\varepsilon]+0\varepsilon$
$\ydiagram{2,2}$	$(2,0)+(0,2)$	$\partial^4\phi^4$	$[8-2\varepsilon]+\frac{7}{9}\varepsilon$
$\ydiagram{4,2}$	$(3,1)+(1,3)$	$\partial^6\phi^3$	$[9-\frac{3}{2}\varepsilon]+0\varepsilon$
$\ydiagram{3,2}$	$(\frac{5}{2},\frac{1}{2})+(\frac{1}{2},\frac{5}{2})$	$\partial^5\phi^4$	$[9-2\varepsilon]+\frac{2}{9}\varepsilon$
$\ydiagram{2,1}$	$(\frac{3}{2},\frac{1}{2})+(\frac{1}{2},\frac{3}{2})$	$\partial^5\phi^4$	$[9-2\varepsilon]+\frac{1}{2}\varepsilon$
$\ydiagram{4,1}$	$(\frac{5}{2},\frac{3}{2})+(\frac{3}{2},\frac{5}{2})$	$\partial^5\phi^4$	$[9-2\varepsilon]+\frac{13}{18}\varepsilon$
$\ydiagram{2,2}$	$(2,0)+(0,2)$	$\partial^4\phi^5$	$[9-\frac{5}{2}\varepsilon]+\frac{17}{9}\varepsilon$
$\ydiagram{5,2}$	$(\frac{7}{2},\frac{3}{2})+(\frac{3}{2},\frac{7}{2})$	$\partial^7\phi^3$	$[10-\frac{3}{2}\varepsilon]+0\varepsilon$
$\ydiagram{6,1}$	$(\frac{7}{2},\frac{5}{2})+(\frac{5}{2},\frac{7}{2})$	$\partial^7\phi^3$	$[10-\frac{3}{2}\varepsilon]+0\varepsilon$
$\ydiagram{3,1}$	$(2,1)+(1,2)$	$\partial^6\phi^4$	$[10-2\varepsilon]+\frac{28}{45}\varepsilon$
$\ydiagram{4,2}$	$(3,1)+(1,3)$	$\partial^6\phi^4$	$[10-2\varepsilon]+\frac{37-\sqrt{649}}{90}\varepsilon$
$\ydiagram{4,2}$	$(3,1)+(1,3)$	$\partial^6\phi^4$	$[10-2\varepsilon]+\frac{37+\sqrt{649}}{90}\varepsilon$
$\ydiagram{5,1}$	$(3,2)+(2,3)$	$\partial^6\phi^4$	$[10-2\varepsilon]+\frac{13}{45}\varepsilon$
$\ydiagram{2,1}$	$(\frac{3}{2},\frac{1}{2})+(\frac{1}{2},\frac{3}{2})$	$\partial^5\phi^5$	$[10-\frac{5}{2}\varepsilon]+\frac{14}{9}\varepsilon$
$\ydiagram{4,1}$	$(\frac{5}{2},\frac{3}{2})+(\frac{3}{2},\frac{5}{2})$	$\partial^5\phi^5$	$[10-\frac{5}{2}\varepsilon]+\frac{16}{9}\varepsilon$
$\ydiagram{3,2}$	$(\frac{5}{2},\frac{1}{2})+(\frac{1}{2},\frac{5}{2})$	$\partial^5\phi^5$	$[10-\frac{5}{2}\varepsilon]+\frac{23}{18}\varepsilon$
$\ydiagram{2,2}$	$(2,0)+(0,2)$	$\partial^4\phi^6$	$[10-3\varepsilon]+\frac{10}{3}\varepsilon$

Reference [95] gives the family of operators with k fields and $\ydiagram{2,2}$ Lorentz representation, for $k \geqslant 3$. For this family, $\Delta = k[1-\frac{\varepsilon}{2}] + \frac{1}{3}(1/2k(k-1) - 2/3k - 1)\varepsilon + O(\varepsilon^2)$.

3.2.1 Three-row Lorentz Young tableaux

We only comment on operators in the representation given by the Lorentz Young tableau $y_{1,1,1} = \ydiagram{1,1,1}$, since they become parity-odd scalars in three dimensions. For $N=1$ and $d=3$, the theory of a free scalar field has an operator the form $\epsilon^{\mu\nu\rho}\partial_\mu\partial_\nu\partial_\rho\Box^3\phi^4$.

For the interacting theory, the corresponding operator is the leading operator in the $y_{1,1,1}$ Lorentz representation, which has dimension $[13-2\varepsilon]+0\varepsilon+O(\varepsilon^2)$. In [100], the numerical conformal bootstrap studied the four-point function of stress-tensors in three dimensions. A general upper bound for the parity-odd scalar was found: $\Delta_{\mathcal{O}_{\mathrm{odd,min}}} \lesssim 11.5$. When inserting some input to single out the Ising model, a somewhat lower bound was found: $\Delta_{\mathcal{O}_{\mathrm{odd,min}}} < 11.2$. Reference [23] found an upper bound $\Delta_{\mathcal{O}_{\mathrm{odd,min}}} \leqslant 10.9293$ within the Ising bootstrap island.

3.3 OPE coefficients

See tables 16, 17 and 18 for OPE coefficients in the $\phi \times \phi$, $\phi \times \phi^2$ and $\phi^2 \times \phi^2$ OPEs.

Several OPE coefficients involving the stress tensor were computed in [24].

Table 16: Squared OPE coefficients in $\phi \times \phi$ for operators for $N = 1$. Numerical values are taken from [96] and denote non-squared OPE coefficients.

$\phi, \phi, \mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\lambda^2_{\phi\phi\mathcal{O}}(\varepsilon)$	$\lambda^2_{\phi\phi\mathcal{O}}(\varepsilon)$	$\lambda^{3d}_{\phi\phi\mathcal{O}}$
$\phi, \phi,$ `Op[E,0,1]`	ϕ^2	2	ε^3[101]ε^4(num)[102]	1.0518537(41)
$\phi, \phi,$ `Op[E,0,2]`	ϕ^4	$\frac{1}{54}\varepsilon^2$	ε^2[103]ε^3[102]	0.053012(55)*
$\phi, \phi,$ `Op[E,0,3]`	ϕ^6	$\frac{5}{104976}\varepsilon^4$	ε^4[104]	0.0007338(31)
$\phi, \phi,$ `Op[E,0,4]`	$\Box^2\phi^4$			0.000162(12)
$\phi, \phi,$ `Op[E,2,1]`	$T^{\mu\nu}$	$\frac{1}{3}$	ε^3[101]ε^4[105]	0.32613776(45)†
$\phi, \phi,$ `Op[E,2,2]`	$\partial^2\phi^4$	$\frac{1}{1440}\varepsilon^2$	ε^2[105]ε^3[7]	0.0105745(42)‡
$\phi, \phi,$ `Op[E,2,3]`	$\partial^2\Box\phi^4$	$\frac{\varepsilon^4}{2592000}$	ε^4[7]	0.0004773(62)
$\phi, \phi,$ `Op[E,4,1]`	$C^{\mu\nu\rho\sigma}$	$\frac{1}{35}$	ε^3[101]ε^4[105]	0.069076(43)
$\phi, \phi,$ `Op[E,4,2]`	$\partial^4\phi^4$	$\frac{1}{419580}\varepsilon^2$	ε^2[1, 105]	0.0019552(12)
$\phi, \phi,$ `Op[E,4,3]`	$\partial^4\phi^4$	$\frac{1}{23976}\varepsilon^2$	ε^2[1, 105]	0.00237745(44)
$\phi, \phi,$ `Op[E,6,1]`	$\mathcal{J}_6$	$\frac{1}{462}$	ε^3[101]ε^4[105]	0.0157416(41)
$\phi, \phi,$ `Op[E,6,2]`	$\partial^6\phi^4$	$O(\varepsilon^2)$	ε^2[1, 105]	
$\phi, \phi,$ `Op[E,6,3]`	$\partial^6\phi^4$	$O(\varepsilon^2)$	ε^2[1, 105]	
$\phi, \phi,$ `Op[E,6,4]`	$\partial^6\phi^4$	$O(\varepsilon^2)$	ε^2[1, 105]	
$\phi, \phi,$ `Op[E,8,1]`	$\mathcal{J}_8$	$\frac{1}{6435}$	ε^3[101]ε^4[105]	0.0036850(54)
$\phi, \phi,$ `Op[E,10,1]`	$\mathcal{J}_{10}$	$\frac{1}{92378}$	ε^3[101]ε^4[105]	0.00087562(13)

*A value with rigorous error bars, 0.05304(16), was given in [78].

†Related to the central charge C_T.

‡A value with rigorous error bars, 0.01054(10), was given in [78].

Table 17: Squared OPE coefficients in $\phi \times \phi^2$ for operators for $N = 1$. Numerical values are from [96] and denote non-squared OPE coefficients.

$\phi, \phi^2, \mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\lambda^2_{\phi\phi^2\mathcal{O}}(\varepsilon)$	$\lambda^2_{\phi\phi^2\mathcal{O}}(\varepsilon)$	$\lambda^{3d}_{\phi\phi^2\mathcal{O}}$
$\phi, \phi^2,$ `Op[0,0,1]`	ϕ	2	ε^3[101]ε^4(num)[102]	1.0518537(41)
$\phi, \phi^2,$ `Op[0,0,2]`	ϕ^5	$\frac{5}{108}\varepsilon^2$	ε^2[104]	0.057235(20)*
$\phi, \phi^2,$ `Op[0,2,1]`	$\partial^2\phi^3$	$\frac{1}{2}$	ε^1[7]	0.38915941(81)
$\phi, \phi^2,$ `Op[0,2,2]`	$\partial^2\phi^5$	$O(\varepsilon^2)$		0.017413(73)
$\phi, \phi^2,$ `Op[0,3,1]`	$\partial^3\phi^3$	$\frac{2}{35}$	ε^1[7]	0.1385(34)
$\phi, \phi^2,$ `Op[0,4,1]`	$\partial^4\phi^3$	$\frac{1}{18}$	ε^1[7]	0.1077052(16)
$\phi, \phi^2,$ `Op[0,5,1]`	$\partial^5\phi^3$	$\frac{2}{231}$	ε^1[7]	0.04191549(88)
$\phi, \phi^2,$ `Op[0,6,1]`	$\partial^6\phi^3$	$O(\varepsilon^2)$		
$\phi, \phi^2,$ `Op[0,6,2]`	$\partial^6\phi^3$	$\frac{3}{572}$	ε^1[7]	
$\phi, \phi^2,$ `Op[0,7,1]`	$\partial^7\phi^3$	$\frac{2}{2145}$	ε^1[7]	0.01161255(13)
$\phi, \phi^2,$ `Op[0,8,1]`	$\partial^8\phi^3$	$O(\varepsilon^2)$		
$\phi, \phi^2,$ `Op[0,8,2]`	$\partial^8\phi^3$	$\frac{1}{2210}$	ε^1[7]	
$\phi, \phi^2,$ `Op[0,9,1]`	$\partial^9\phi^3$	$O(\varepsilon^2)$		
$\phi, \phi^2,$ `Op[0,9,2]`	$\partial^9\phi^3$	$\frac{4}{46189}$	ε^1[7]	

*A value with rigorous error bars, 0.0565(15), was given in [78].

Table 18: Squared OPE coefficients in $\phi^2 \times \phi^2$ for operators for $N = 1$. Numerical values are from [96] and denote non-squared OPE coefficients.

$\phi^2, \phi^2, \mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\lambda^2_{\phi^2\phi^2\mathcal{O}}(\varepsilon)$	$\lambda^2_{\phi^2\phi^2\mathcal{O}}(\varepsilon)$	$\lambda^{3d}_{\phi^2\phi^2\mathcal{O}}$
$\phi^2, \phi^2,$ `Op[E,0,0]`	$\mathbb{1}$	1	exact	1
$\phi^2, \phi^2,$ `Op[E,0,1]`	ϕ^2	8	ε^1[1, 106]	1.532435(19)
$\phi^2, \phi^2,$ `Op[E,0,2]`	ϕ^4	6	ε^1[1, 106]	1.5360(16)*
$\phi^2, \phi^2,$ `Op[E,0,3]`	ϕ^6			0.1279(17)
$\phi^2, \phi^2,$ `Op[E,0,4]`	$\Box^2\phi^4$	$\frac{1}{6}$	ε^1[1, 106]	0.1874(31)
$\phi^2, \phi^2,$ `Op[E,2,1]`	$T^{\mu\nu}$	$\frac{4}{3}$	ε^4[105]	0.8891471(40)
$\phi^2, \phi^2,$ `Op[E,2,2]`	$\partial^2\phi^4$	$\frac{8}{5}$	ε^1[1, 106]	0.69023(49)†
$\phi^2, \phi^2,$ `Op[E,2,3]`	$\partial^2\Box\phi^4$	$\frac{1}{5}$	ε^1[1, 106]	0.21882(73)
$\phi^2, \phi^2,$ `Op[E,2,4]`	$\partial^2\phi^6$			
$\phi^2, \phi^2,$ `Op[E,2,5]`	$\partial^2\Box^2\phi^4$	$\frac{1}{999}$	ε^1[1, 7, 106]	
$\phi^2, \phi^2,$ `Op[E,2,6]`	$\partial^2\Box^2\phi^4$	$\frac{4}{111}$	ε^1[1, 7, 106]	
$\phi^2, \phi^2,$ `Op[E,4,1]`	$C^{\mu\nu\rho\sigma}$	$\frac{4}{35}$	ε^1[1, 106]	0.24792(20)
$\phi^2, \phi^2,$ `Op[E,4,2]`	$\partial^4\phi^4$	$\frac{125}{2331}$	ε^1[1, 106]	−0.110247(54)
$\phi^2, \phi^2,$ `Op[E,4,3]`	$\partial^4\phi^4$	$\frac{8}{37}$	ε^1[1, 106]	0.22975(10)
$\phi^2, \phi^2,$ `Op[E,4,4]`	$\partial^4\Box\phi^4$	$\frac{2091-71\sqrt{697}}{107338}$	ε^1[1, 7, 106]	0.08635(18)
$\phi^2, \phi^2,$ `Op[E,4,5]`	$\partial^4\Box\phi^4$	$\frac{2091+71\sqrt{697}}{107338}$	ε^1[1, 7, 106]	
$\phi^2, \phi^2,$ `Op[E,6,1]`	$\mathcal{J}_6$	$\frac{2}{231}$	ε^1[1, 106]	0.066136(36)
$\phi^2, \phi^2,$ `Op[E,8,1]`	$\mathcal{J}_8$	$\frac{4}{6435}$	ε^1[1, 106]	0.017318(30)
$\phi^2, \phi^2,$ `Op[E,10,1]`	$\mathcal{J}_{10}$	$\frac{2}{46189}$	ε^1[1, 106]	0.0044811(15)

*A value with rigorous error bars, 1.5362(12), was given in [78].
†A value with rigorous error bars, 0.678(15), was given in [78].

3.4 Families of operators for general N

For general N we introduce a number of families $\texttt{ONF}\langle i\rangle$. We express the $1/N$ part of the large N operator dimension in terms of the anomalous dimension of φ: $\Delta_\varphi = \mu-1+\frac{\eta_1/2}{N}+O(N^{-2})$.

$\texttt{ONF1[}k\texttt{]}$ Scalar singlet operators of the form $\varphi_S^{2k} \sim \sigma^k$, $k \geqslant 1$. We have ε^1[107]ε^2[90] and N^{-1}[71, 108]N^{-2}[90],

$$\texttt{DeltaE[ONF1[}k\texttt{]]} = k[2-\varepsilon] + \frac{k(6k+N-4)}{N+8}\varepsilon + \ldots + O(\varepsilon^3), \tag{3.15}$$

$$\texttt{DeltaN[ONF1[}k\texttt{]]} = 2k + \frac{2k(2\mu-1)}{\mu-2}\left(k\mu(2\mu-3)-\mu^2+\mu+2\right)\frac{\eta_1/2}{N} + \ldots + O(1/N^3). \tag{3.16}$$

$\texttt{ONF2[}k\texttt{]}$ Scalar fundamental operators of the form $\varphi_V^{2k+1} \sim \sigma^k\varphi$, $k \geqslant 0$, $k \neq 1$. We have ε^1[107] and N^{-1}[66],

$$\texttt{DeltaE[ONF2[}k\texttt{]]} = (2k+1)[1-\tfrac{\varepsilon}{2}] + \frac{k(6k+N+2)}{N+8}\varepsilon + O(\varepsilon^2), \tag{3.17}$$

$$\texttt{DeltaN[ONF2[}k\texttt{]]} = 2k+\mu-1 + \left(\frac{2k(k-1)\mu(2\mu-3)(2\mu-1)}{2-\mu}+1\right)\frac{\eta_1/2}{N} + O(N^{-2}). \tag{3.18}$$

In fact $\texttt{ONF1}$ and $\texttt{ONF2}$ have the largest order ε anomalous dimensions for operator with $2k$ and $2k+1$ fields respectively [94].

$\texttt{ONF3[}m\texttt{]}$ Scalar completely symmetric traceless tensors of the form $\varphi_{T_m}^m$, $m \geqslant 1$. We have ε^2[109]ε^4[110]ε^5[1, 60, 110]ε^6[111] and N^{-1}[112, 113]N^{-2}[62],

$$\texttt{DeltaE[ONF3[}m\texttt{]]} = m\left[1-\frac{1}{2}\varepsilon\right] + \frac{m(m-1)}{N+8}\varepsilon + \ldots + O(\varepsilon^6), \tag{3.19}$$

$$\texttt{DeltaN[ONF3[}m\texttt{]]} = m(\mu-1) + \frac{m(m\mu+2-2\mu)}{\mu-2}\frac{\eta_1/2}{N} + \ldots + O(N^{-3}). \tag{3.20}$$

$\texttt{ONF4[}l\texttt{]}$ Singlet broken currents of the form $\mathcal{J}_{S,\ell} = [\varphi,\varphi]_{S,0,\ell}$, $\ell \in 2\mathbb{Z}_+$. We have ε^2[92]ε^4[114] and N^{-1}[108]N^{-2}[114],

$$\texttt{DeltaE[ONF4[}l\texttt{]]} = 2-\varepsilon+l + \frac{N+2}{2(N+8)^2}\left(1-\frac{6}{l(l+1)}\right)\varepsilon^2 + \ldots + O(\varepsilon^5), \tag{3.21}$$

$$\texttt{DeltaN[ONF4[}l\texttt{]]} = 2(\mu-1)+l + \left(2 - \frac{2\mu\left((\mu-1)+\frac{\Gamma(2\mu-1)\Gamma(l+1)}{\Gamma(l+2\mu-3)}\right)}{(l+\mu-1)(l+\mu-2)}\right)\frac{\eta_1/2}{N} + \ldots + O(N^{-3}). \tag{3.22}$$

`ONF5[`l`]` Traceless-symmetric broken currents of the form $\mathcal{J}_{T,\ell} = [\varphi,\varphi]_{T,0,\ell}$, $\ell \in 2\mathbb{Z}_+$. We have ε^2[92]ε^3[115]ε^4[65] and N^{-1}[108]N^{-2}[62],

$$\texttt{DeltaE[ONF5[}l\texttt{]]} = 2 - \varepsilon + l + \frac{1}{2(N+8)^2}\left(N + 2 - \frac{2(N+6)}{l(l+1)}\right)\varepsilon^2 + \ldots + O(\varepsilon^5), \quad (3.23)$$

$$\texttt{DeltaN[ONF5[}l\texttt{]]} = 2(\mu-1) + l + \left(2 - \frac{2\mu(\mu-1)}{(l+\mu-1)(l+\mu-2)}\right)\frac{\eta_1/2}{N} + \ldots + O(N^{-3}). \quad (3.24)$$

`ONF6[`l`]` Antisymmetric broken currents of the form $\mathcal{J}_{A,\ell} = [\varphi,\varphi]_{A,0,\ell}$, $\ell \in 2\mathbb{N}+1$. We have ε^3[115]ε^4[65] and N^{-1}[108]N^{-2}[62],

$$\texttt{DeltaE[ONF6[}l\texttt{]]} = 2 - \varepsilon + l + \frac{N+2}{2(N+8)^2}\left(1 - \frac{2}{l(l+1)}\right)\varepsilon^2 + \ldots + O(\varepsilon^5), \quad (3.25)$$

$$\texttt{DeltaN[ONF6[}l\texttt{]]} = 2(\mu-1) + l + \left(2 - \frac{2\mu(\mu-1)}{(l+\mu-1)(l+\mu-2)}\right)\frac{\eta_1/2}{N} + \ldots + O(N^{-3}). \quad (3.26)$$

`ONF7[`l`]` Spinning singlet operators of the form $[\sigma,\sigma]_{0,\ell}$, $\ell \in 2\mathbb{N}$. We have N^{-1}[108],

$$\begin{aligned}\texttt{DeltaN[ONF7[}l\texttt{]]} = 4 + l - \frac{8}{(2-\mu)^2}\Bigg((\mu-1)(2-\mu)(2\mu-1) \quad (3.27)\\ + \frac{\mu(2\mu-3)}{(l+1)(l+2)}\left((\mu-1)(2\mu-3) - \frac{(\mu-l-3)_l}{(\mu)_l}\right)\Bigg)\frac{\eta_1/2}{N} + O(N^{-2}),\end{aligned}$$

where $(a)_k = \frac{\Gamma(a+k)}{\Gamma(a)}$ is the (rising) Pochhammer symbol. There is no simple expression for these operators in the ε-expansion.

`ONF8[`l`]` Spinning fundamental operators of the form $[\varphi,\sigma]_{0,\ell}$, $\ell \geqslant 1$. We have ε^1[94] and N^{-1}[108],

$$\begin{aligned}\texttt{DeltaE[ONF8[}l\texttt{]]} = [3 - \tfrac{3}{2}\varepsilon] + l \quad (3.28)\\ + \frac{4(-1)^l + (N+4)(l+1) + \sqrt{16(l+1)(-1)^l + 48 + 16N + (l+1)^2N^2}}{2(N+8)(l+1)}\varepsilon + O(\varepsilon^2),\end{aligned}$$

$$\begin{aligned}\texttt{DeltaN[ONF8[}l\texttt{]]} = \mu + 1 + l + \Bigg(\frac{4\mu(\mu-1)(2\mu-3)}{(l+1)(l+\mu-1)(2-\mu)} \quad (3.29)\\ - \frac{8\mu^2 - 11\mu + 2}{2-\mu} + \frac{2(-1)^l\Gamma(l+1)\Gamma(\mu+1)}{(2-\mu)\Gamma(l+\mu)}\Bigg)\frac{\eta_1/2}{N} + O(N^{-2}).\end{aligned}$$

`ONF9[`k`]` Scalar singlet operators of the form $\Box\varphi_S^{2k} \sim \Box\sigma^k$, $k \geqslant 2$. We have ε^1[1] and N^{-1}[71],

$$\texttt{DeltaE[ONF9[}k\texttt{]]} = 2 + k(2-\varepsilon) + \frac{6k^2+(N-12)k+4}{N+8}\varepsilon + O(\varepsilon^2), \tag{3.30}$$

$$\begin{aligned}\texttt{DeltaN[ONF9[}k\texttt{]]} = 2k+2+\frac{2(2\mu-1)}{3(2-\mu)}\big(&3k^2\mu(2\mu-3)-k(10\mu^2-\mu-18)\\ &+2\mu^3-\mu^2-7\mu+6\big)\frac{\eta_1/2}{N}+O(N^{-2}).\end{aligned} \tag{3.31}$$

`ONF10[`k`,`m`]` Scalar operators of the form $\varphi_{T_m}^{2k+m} \sim \sigma^k\varphi_{T_m}^m$, $k, m \geqslant 0$. We have ε^1[107] and N^{-1}[66],

$$\texttt{DeltaE[ONF10[}k\texttt{,}m\texttt{]]} = (2k+m)[1-\tfrac{\varepsilon}{2}] + \frac{6k^2+k(6m+N-4)+m(m-1)}{N+8}\varepsilon + O(\varepsilon^2), \tag{3.32}$$

$$\begin{aligned}\texttt{DeltaN[ONF10[}k\texttt{,}m\texttt{]]} = 2k+m(\mu-1)+\frac{1}{2-\mu}\big[&2k^2\mu(2\mu-1)(2\mu-3)+m(m\mu+2-2\mu)\\ &-2k(2\mu-1)(2\mu^2-\mu-2-2m(\mu-1))\big]\frac{\eta_1/2}{N}+O(N^{-2}).\end{aligned} \tag{3.33}$$

For $m=0$ the family reduces to `ONF1[`k`]`, for $m=1$ to `ONF2[`k`]` and for $k=0$ to `ONF3[`m`]`.

`ONF11[`m`,`l`]` Spin-l operators of the form $\partial^l\varphi_{Y_{m,l}}^{m+l}$ transforming in the O(N) representation $Y_{m,l}$. We have ε^1[1] and N^{-1}[113].

$$\texttt{DeltaE[ONF11[}m\texttt{,}l\texttt{]]} = (m+l)[1-\tfrac{\varepsilon}{2}] + l + \frac{3m^2+3ml-3m+l^2-4l}{3(N+8)}\varepsilon + O(\varepsilon^2), \tag{3.34}$$

$$\begin{aligned}\texttt{DeltaN[ONF11[}m\texttt{,}l\texttt{]]} &= (m+l)(\mu-1)+l\\ &+\left[(m+l)(m+l-2)-4(l-1)\left(\tfrac{m}{2\mu}+\tfrac{l}{2\mu+2}\right)\right]\frac{\mu}{2-\mu}\frac{\eta_1/2}{N}+O(N^{-2}).\end{aligned} \tag{3.35}$$

For $l=0$ the family reduces to `ONF3[`m`]`. Note that $Y_{m,0}=T_m$, and $Y_{m,1}=H_{m+1}$, and that $Y_{2,2}=B_4$. Note also that operators with $m+l$ fields in the O(N) representation $Y_{m,l}$ and with no contracted derivatives must have spin $\geqslant l$.

`ONF12[`m`]` Spin-2 operators of the form $\partial^2\varphi_{T_m}^m$ transforming in the rank-m traceless-symmetric $O(N)$ representation, $m \geqslant 2$. We have ε^2[2, 99] and N^{-1}[2, 116]

$$\texttt{DeltaE[ONF12[}m\texttt{]]} = m[1-\tfrac{\varepsilon}{2}] + 2 + \frac{3m^2-5m-2}{3(N+8)}\varepsilon + \ldots + O(\varepsilon^3), \tag{3.36}$$

$$\texttt{DeltaN[ONF11[}m\texttt{]]} = m(\mu-1)+2+\frac{m^2\mu(\mu+1)-2m(\mu^2+2\mu-3)-4}{(\mu+1)(2-\mu)}\frac{\eta_1/2}{N}+O(N^{-2}). \tag{3.37}$$

For $l = 0$ the family reduces to `ONF3[`m`]`. Note that $Y_{m,0} = T_m$, and $Y_{m,1} = H_{m+1}$, and that $Y_{2,2} = B_4$. Note also that operators with $m + l$ fields in the $\mathrm{O}(N)$ representation $Y_{m,l}$ and with no contracted derivatives must have spin $\geqslant l$.

Apart from the eleven twelve families of operators for generic N listed above, [94] gives three additional families in the ε-expansion. These families are only defined by the fact that they acquire an order ε anomalous dimension, which singles them out from additional operators of the same type. This property is not visible in the large N expansion, whence we do not include them in our list of families.

- $\partial^\ell \varphi^3_V$ with non-zero order ε dimension ε^1[94]. This is the sister family to `ONF8[`ℓ`]`. Dimension given by (3.28), but with a minus sign in front of the square root.
- $\partial^\ell \varphi^3_{T_3}$ with non-zero order ε dimension ε^1[94]: , $\ell \neq 1$. By computing many of these operator dimensions using [99], we have guessed a general two-loop formula [2]:

$$\begin{aligned}\Delta_\ell = {}& [3 - \frac{3}{2}\varepsilon] + \ell + \frac{2}{N+8}\left(1 + \frac{2(-1)^\ell}{\ell+1}\right)\varepsilon + \varepsilon^2\Bigg[- \frac{N^2 - 46N - 192}{4(N+8)^3} \\ & + \frac{4(N+6)(\ell^3 - (\ell+1)^3 S_1(\ell)) + 2\ell^2(5N+26) + 4\ell(3N+14) + 6N + 60}{(N+8)^2(\ell+3)(\ell+1)^3(\ell-1)} \\ & - 2(-1)^\ell\Bigg(\frac{\ell^3(3N^2+28N+92)+\ell^2(7N^2+54N+164)-\ell(2N^2+18N+76)+48N+204}{(N+8)^3(\ell+3)(\ell+1)^2(\ell-1)} \\ & \qquad - \frac{(N+8)(\ell^2+2\ell)+N}{(N+8)^2(\ell+3)(\ell+1)(\ell-1)} S_1(\ell)\Bigg)\Bigg] + O(\varepsilon^3)\end{aligned} \tag{3.38}$$

- $\partial^\ell \varphi^3_{H_3}$ with non-zero order ε dimension ε^1[94]: $[3 - \frac{3}{2}\varepsilon] + \ell + \frac{1}{N+8}\left(2 - \frac{2(-1)^\ell}{\ell+1}\right)\varepsilon + O(\varepsilon^2)$, $\ell \neq 0$.

Moreover, in [66], a closed-form expression at large N is given for the anomalous dimension of spin-one operators of the form $\partial\sigma^k\varphi$.

Table 19: Singlet scalar operators for $N > 1$. $N^0_{\dots}$ denotes $N^0 + O(N^{-1})$, where there is an exact expression.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[S,0,1]`	φ_S^2	σ	$[2-\varepsilon] + \frac{N+2}{N+8}\varepsilon$	ε^7[40]	N^{-2}[49]	1_2, $10_{1,0}$
`Op[S,0,2]`	φ_S^4	σ^2	$[4-2\varepsilon] + 2\varepsilon$	ε^7[40]	N^{-2}[51]	1_2, 7_0, $10_{2,0}$
`Op[S,0,3]`	$\Box\varphi_S^4$	$\Box\sigma^2$	$[6-2\varepsilon] + \frac{2N+4}{N+8}\varepsilon$	ε^4[4]ε^5[5]	N^{-1}[71]	9_2
`Op[S,0,4]`	φ_S^6	σ^3	$[6-3\varepsilon] + \frac{3N+42}{N+8}$	ε^2[90]ε^4[4]ε^5[5]	N^{-2}[90]	1_3, $10_{3,0}$
`Op[S,0,5]`	$\Box^2\varphi_S^4$	$\Box^2\varphi_S^4$	$[8-2\varepsilon] + 0\varepsilon N^0_{\dots}$	ε^3[4]ε^4[5]		
`Op[S,0,6]`	$\Box^2\varphi_S^4$	$\Box^2\sigma^2$	$[8-2\varepsilon] + 2\varepsilon N^0_{\dots}$	ε^3[4]ε^4[5]		
`Op[S,0,7]`	$\Box\varphi_S^6$	$\Box\sigma^3$	$[8-3\varepsilon] + \frac{3N+22}{N+8}\varepsilon$	ε^3[4]ε^4[5]	N^{-1}[71]	9_3
`Op[S,0,8]`	φ_S^8	σ^4	$[8-4\varepsilon] + \frac{4N+80}{N+8}\varepsilon$	ε^2[90]ε^3[4]ε^4[5]	N^{-2}[90]	1_4, $10_{4,0}$
`Op[S,0,9]`	$\Box^3\varphi_S^4$		$[10-2\varepsilon] + 0\varepsilon N^0_{\dots}$	ε^2[5]		
`Op[S,0,10]`	$\Box^2\varphi_S^6$		$[10-3\varepsilon] + \varepsilon N^0_{\dots}$	ε^2[5]		
`Op[S,0,11]`	$\Box^2\varphi_S^6$		$[10-3\varepsilon] + 2\varepsilon N^0_{\dots}$	ε^2[5]		
`Op[S,0,12]`	$\Box^2\varphi_S^6$		$[10-3\varepsilon] + 3\varepsilon N^0_{\dots}$	ε^2[5]		
`Op[S,0,13]`	$\Box^3\varphi_S^4$		$[10-2\varepsilon] + 2\varepsilon N^0_{\dots}$	ε^2[5]		
`Op[S,0,14]`	$\Box^2\varphi_S^6$		$[10-3\varepsilon] + 3\varepsilon N^0_{\dots}$	ε^2[5]		
`Op[S,0,15]`	$\Box\varphi_S^8$	$\Box\sigma^4$	$[10-4\varepsilon] + \frac{4N+13}{N+8}\varepsilon$	ε^2[5]	N^{-1}[71]	9_4
`Op[S,0,16]`	φ_S^{10}	σ^5	$[10-5\varepsilon] + \frac{5N+130}{N+8}\varepsilon$	ε^2[90]	N^{-2}[90]	1_5, $10_{5,0}$

3.4.1 Spinning singlet operators

Table 20: Singlet spinning operators for $N > 1$.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[S,1,1]`	$\partial\Box\varphi_S^6$	$\partial\Box\sigma^3$	$[9-3\varepsilon]+\frac{9N+58}{3(N+8)}\varepsilon$			
`Op[S,2,1]`	$T^{\mu\nu}$	$T^{\mu\nu}$	$[4-\varepsilon]$	exact	exact	4_2
`Op[S,2,2]`	$\partial^2\varphi_S^4$	$[\varphi,\varphi]_{S,1,2}$	$[6-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^5[5]		
`Op[S,2,3]`	$\partial^2\varphi_S^4$	$[\sigma,\sigma]_{0,2}$	$[6-2\varepsilon]+2\varepsilon N^0_{\dots}$	ε^5[5]	N^{-1}[108]	7_2
`Op[S,2,4]`	$\partial^2\Box\varphi_S^4$		$[8-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[S,2,5]`	$\partial^2\Box\varphi_S^4$		$[8-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[S,2,6]`	$\partial^2\varphi_S^6$		$[8-3\varepsilon]+2\varepsilon N^0_{\dots}$	ε^1		
`Op[S,2,7]`	$\partial^2\Box\varphi_S^4$		$[8-2\varepsilon]+2\varepsilon N^0_{\dots}$	ε^1		
`Op[S,2,8]`	$\partial^2\varphi_S^6$		$[8-3\varepsilon]+3\varepsilon N^0_{\dots}$	ε^1		
`Op[S,3,1]`	$\partial^3\Box\varphi_S^4$		$[9-2\varepsilon]+\frac{n+2}{n+8}\varepsilon$	ε^1		
`Op[S,3,2]`	$\partial^3\varphi_S^6$		$[9-3\varepsilon]+2\varepsilon N^0_{\dots}$	ε^1		
`Op[S,3,3]`	$\partial^3\varphi_S^6$		$[9-3\varepsilon]+3\varepsilon N^0_{\dots}$	ε^1		
`Op[S,4,1]`	$\mathcal{J}_{S,4}$	$\mathcal{J}_{S,4}$	$[6-\varepsilon]$	ε^4[114]	N^{-2}[114]	4_4
`Op[S,4,2]`	$\partial^4\varphi_S^4$	$\partial^4\varphi_S^4$	$[8-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[S,4,3]`	$\partial^4\varphi_S^4$		$[8-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[S,4,4]`	$\partial^4\varphi_S^4$		$[8-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[S,4,5]`	$\partial^4\varphi_S^4$	$[\sigma,\sigma]_{0,4}$	$[8-2\varepsilon]+2\varepsilon N^0_{\dots}$	ε^1	N^{-1}[108]	7_4
`Op[S,5,1]`	$\partial^5\varphi_S^4$		$[9-2\varepsilon]+\frac{N+2}{N+8}\varepsilon$	ε^1		
`Op[S,6,1]`	$\mathcal{J}_{S,6}$	$\mathcal{J}_{S,6}$	$[8-\varepsilon]$	ε^4[114]	N^{-2}[114]	4_6
`Op[S,6,2]`	$\partial^6\varphi_S^4$	$\partial^6\varphi_S^4$	$[10-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[S,6,3]`	$\partial^6\varphi_S^4$	$\partial^6\varphi_S^4$	$[10-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[S,6,4]`	$\partial^6\varphi_S^4$		$[10-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[S,6,5]`	$\partial^6\varphi_S^4$		$[10-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[S,6,6]`	$\partial^6\varphi_S^4$		$[10-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[S,6,7]`	$\partial^6\varphi_S^4$	$[\sigma,\sigma]_{0,6}$	$[10-2\varepsilon]+2\varepsilon N^0_{\dots}$	ε^1	N^{-1}[108]	7_6
`Op[S,8,1]`	$\mathcal{J}_{S,8}$	$\mathcal{J}_{S,8}$	$[10-\varepsilon]$	ε^4[114]	N^{-2}[114]	4_8
`Op[S,10,1]`	$\mathcal{J}_{S,10}$	$\mathcal{J}_{S,10}$	$[12-\varepsilon]$	ε^4[114]	N^{-2}[114]	4_{10}

Table 21: Operators in the vector representation V.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[V,0,1]`	φ	φ	$[1-\frac{1}{2}\varepsilon]+0\varepsilon$	ε^8[41]	N^{-3}[45]*	$2_0, 3_1, 11_{1,0}, 10_{0,1}$
`Op[V,0,2]`	φ^5_V	$\sigma^2\varphi$	$[5-\frac{5}{2}\varepsilon]+\frac{2N+28}{N+8}\varepsilon$	ε^5[5]	N^{-1}[66]	2_2, $10_{2,1}$
`Op[V,0,3]`	$\Box\varphi^5_V$	$\Box\sigma^2\varphi$	$[7-\frac{5}{2}\varepsilon]+\frac{2N+12}{N+8}\varepsilon$	ε^1		
`Op[V,0,4]`	φ^7_V	$\sigma^3\varphi$	$[7-\frac{7}{2}\varepsilon]+\frac{3N+60}{N+8}\varepsilon$	ε^1	N^{-1}[66]	2_3, $10_{3,1}$
`Op[V,1,1]`	$\partial\varphi^3_V$	$\partial\sigma\varphi$	$[4-\frac{3}{2}\varepsilon]+\frac{N+2}{N+8}\varepsilon$	ε^5[5]	N^{-1}[69]	8_1, 12_1
`Op[V,1,2]`	$\partial\varphi^5_V$	$\partial\sigma^2\varphi$	$[6-\frac{5}{2}\varepsilon]+\frac{2N+18}{N+8}\varepsilon$	ε^5[5]	N^{-1}[66]	
`Op[V,2,1]`	$\partial^2\varphi^3_V$		$[5-\frac{3}{2}\varepsilon]+0\varepsilon N^0_{\dots}$	ε^5[5]		
`Op[V,2,2]`	$\partial^2\varphi^3_V$	$\partial^2\sigma\varphi$	$[5-\frac{3}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^5[5]	N^{-1}[108]	8_2
`Op[V,2,3]`	$\partial^2\varphi^5_V$		$[7-\frac{5}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[V,2,4]`	$\partial^2\varphi^5_V$		$[7-\frac{5}{2}\varepsilon]+2\varepsilon N^0_{\dots}$	ε^1		
`Op[V,2,5]`	$\partial^2\varphi^5_V$		$[7-\frac{5}{2}\varepsilon]+2\varepsilon N^0_{\dots}$	ε^1		
`Op[V,3,1]`	$\partial^3\varphi^3_V$		$[6-\frac{3}{2}\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[V,3,2]`	$\partial^3\varphi^3_V$	$\partial^3\sigma\varphi$	$[6-\frac{3}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1	N^{-1}[108]	8_3
`Op[V,4,1]`	$\partial^4\varphi^3_V$		$[7-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[V,4,2]`	$\partial^4\varphi^3_V$		$[7-\frac{3}{2}\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[V,4,3]`	$\partial^4\varphi^3_V$	$\partial^4\sigma\varphi$	$[7-\frac{3}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1	N^{-1}[108]	8_4
`Op[V,5,1]`	$\partial^5\varphi^3_V$		$[8-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[V,5,2]`	$\partial^5\varphi^3_V$		$[8-\frac{3}{2}\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[V,5,3]`	$\partial^5\varphi^3_V$	$\partial^5\sigma\varphi$	$[8-\frac{3}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1	N^{-1}[108]	8_5
`Op[V,6,1]`	$\partial^6\varphi^3_V$		$[9-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[V,6,2]`	$\partial^6\varphi^3_V$		$[9-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[V,6,3]`	$\partial^6\varphi^3_V$		$[9-\frac{3}{2}\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[V,6,4]`	$\partial^6\varphi^3_V$	$\partial^6\sigma\varphi$	$[9-\frac{3}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1	N^{-1}[108]	8_6
`Op[V,7,1]`	$\partial^7\varphi^3_V$		$[10-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[V,7,2]`	$\partial^7\varphi^3_V$		$[10-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[V,7,3]`	$\partial^7\varphi^3_V$		$[10-\frac{3}{2}\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[V,7,4]`	$\partial^7\varphi^3_V$		$[10-\frac{3}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1	N^{-1}[108]	8_7

* As noted in [71], the expression in [45] contains a misprint in equation (22).

Table 22: Operators in the T representation. Note the double eigenvalue at `Op[T,0,7]` and `Op[T,0,8]`, lifted at two-loop.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[T,0,1]`	φ_T^2	φ_T^2	$[2-\varepsilon]+\frac{2}{N+8}\varepsilon$	ε^6[56]	N^{-2}[57]	3_2, $10_{0,2}$, $11_{2,0}$
`Op[T,0,2]`	φ_T^4	$\sigma\varphi_T^2$	$[4-2\varepsilon]+\frac{N+16}{N+8}\varepsilon$	ε^6[60]	N^{-1}[50]	$10_{1,2}$
`Op[T,0,3]`	$\Box\varphi_T^4$		$[6-2\varepsilon]+\frac{N+4}{N+8}\varepsilon$	ε^5[5]		
`Op[T,0,4]`	φ_T^6	$\sigma^2\varphi_T^2$	$[6-3\varepsilon]+\frac{2N+42}{N+8}\varepsilon$	ε^5[5]	N^{-1}[66]	$10_{2,2}$
`Op[T,0,5]`	$\Box^2\varphi_T^4$		$[8-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^2[5]		
`Op[T,0,6]`	$\Box^2\varphi_T^4$		$[8-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^2[5]		
`Op[T,0,7]`	$\Box\varphi_T^6$		$[8-3\varepsilon]+\frac{2N+22}{N+8}\varepsilon$	ε^2[5]		
`Op[T,0,8]`	$\Box\varphi_T^6$		$[8-3\varepsilon]+\frac{2N+22}{N+8}\varepsilon$	ε^2[5]		
`Op[T,0,9]`	φ_T^8	$\sigma^3\varphi_T^2$	$[8-4\varepsilon]+\frac{3N+80}{N+8}\varepsilon$	ε^2[5]	N^{-1}[66]	$10_{3,2}$
`Op[T,1,1]`	$\partial\Box\varphi_T^4$		$[7-2\varepsilon]+\frac{3N+16}{3(N+8)}\varepsilon$	ε^1		
`Op[T,1,2]`	$\partial\varphi_T^6$		$[7-3\varepsilon]+\frac{2N+30}{N+8}\varepsilon$	ε^1		
`Op[T,2,1]`	$\mathcal{J}_{T,2}$	$\mathcal{J}_{T,2}$	$[4-\varepsilon]+0\varepsilon$	ε^4[65]ε^5[5]	N^{-2}[62]	5_2,12_2
`Op[T,2,2]`	$\partial^2\varphi_T^4$	$\partial^2\varphi_T^4$	$[6-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^5[5]		
`Op[T,2,3]`	$\partial^2\varphi_T^4$		$[6-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^5[5]		
`Op[T,2,4]`	$\partial^2\varphi_T^4$		$[6-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^5[5]		
`Op[T,3,1]`	$\partial^3\varphi_T^4$		$[7-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[T,3,2]`	$\partial^3\varphi_T^4$		$[7-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[T,4,1]`	$\mathcal{J}_{T,4}$	$\mathcal{J}_{T,4}$	$[6-\varepsilon]+0\varepsilon$	ε^4[65]	N^{-2}[62]	5_4
`Op[T,6,1]`	$\mathcal{J}_{T,6}$	$\mathcal{J}_{T,6}$	$[8-\varepsilon]+0\varepsilon$	ε^4[65]	N^{-2}[62]	5_6

Table 23: Operators in the A representation.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[A,1,1]`	J^{μ}_{ij}	J^{μ}_{ij}	$[3-\varepsilon]$	exact	exact	6_1
`Op[A,1,2]`	$\partial\varphi^4_A$		$[5-2\varepsilon]+\frac{N+10}{N+8}\varepsilon$	ε^5[5]		
`Op[A,1,3]`	$\partial\Box\varphi^4_A$		$[7-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[A,1,4]`	$\partial\Box\varphi^4_A$		$[7-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[A,1,5]`	$\partial\varphi^6_A$		$[7-3\varepsilon]+\frac{2N+32}{N+8}\varepsilon$	ε^1		
`Op[A,3,1]`	$\mathcal{J}_{A,3}$	$\mathcal{J}_{A,3}$	$[5-\varepsilon]$	ε^4[65]	N^{-2}[62]	6_3
`Op[A,3,2]`	$\partial^3\varphi^4_A$		$[7-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[A,3,3]`	$\partial^3\varphi^4_A$		$[7-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[A,3,4]`	$\partial^3\varphi^4_A$		$[7-2\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[A,5,1]`	$\mathcal{J}_{A,5}$	$\mathcal{J}_{A,5}$	$[7-\varepsilon]$	ε^4[65]	N^{-2}[62]	6_5

Table 24: Some operators in the rank 3 traceless-symmetric representation.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[Tm[3],0,1]`	$\varphi^3_{T_3}$	$\varphi^3_{T_3}$	$[3-\frac{3}{2}\varepsilon]+\frac{6}{N+8}\varepsilon$	ε^6[60]	N^{-2}[62]	3_3, $10_{0,3}$, $11_{3,0}$
`Op[Tm[3],0,2]`	$\varphi^5_{T_3}$	$\sigma\varphi^3_{T_3}$	$[5-\frac{5}{2}\varepsilon]+\frac{N+26}{N+8}\varepsilon$	ε^5[5]	N^{-1}[66]	$10_{1,3}$
`Op[Tm[3],0,3]`	$\Box\varphi^5_{T_3}$	$\sigma\Box\varphi^5_{T_3}$	$[7-\frac{5}{2}\varepsilon]+\frac{N+10}{N+8}\varepsilon$	ε^1		
`Op[Tm[3],0,4]`	$\varphi^7_{T_3}$	$\sigma^2\varphi^3_{T_3}$	$[7-\frac{7}{2}\varepsilon]+\frac{2N+58}{N+8}\varepsilon$	ε^1	N^{-1}[66]	$10_{2,3}$
`Op[Tm[3],1,1]`	$\partial\varphi^5_{T_3}$	$\partial\sigma\varphi^3_{T_3}$	$[6-\frac{5}{2}\varepsilon]+\frac{N+16}{N+8}\varepsilon$	ε^5[5]		
`Op[Tm[3],2,1]`	$\partial^2\varphi^3_{T_3}$	$\partial^2\varphi^3_{T_3}$	$[5-\frac{3}{2}\varepsilon]+\frac{10}{3(N+8)}\varepsilon$	ε^2[99]ε^5[5]	N^{-1}[1, 116]	12_3
`Op[Tm[3],2,2]`	$\partial^2\varphi^5_{T_3}$	$\partial^2\varphi^3_{T_5}$	$[7-\frac{5}{2}\varepsilon]+0\varepsilon N^0_{\dots}$	ε^1		
`Op[Tm[3],2,3]`	$\partial^2\varphi^5_{T_3}$	$\partial^2\sigma\varphi^3_{T_3}$	$[7-\frac{5}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[Tm[3],2,4]`	$\partial^2\varphi^5_{T_3}$	$\partial^2\sigma\varphi^3_{T_3}$	$[7-\frac{5}{2}\varepsilon]+\varepsilon N^0_{\dots}$	ε^1		
`Op[Tm[3],3,1]`	$\partial^3\varphi^3_{T_3}$	$\partial^3\varphi^3_{T_3}$	$[6-\frac{3}{2}\varepsilon]+\frac{1}{N+8}\varepsilon$	ε^2[99]	N^{-1}[1, 116]	
`Op[Tm[3],4,1]`	$\partial^4\varphi^3_{T_3}$	$\partial^4\varphi^3_{T_3}$	$[7-\frac{3}{2}\varepsilon]+\frac{14}{5(N+8)}\varepsilon$	ε^2[99]	N^{-1}[1, 116]	
`Op[Tm[3],5,1]`	$\partial^5\varphi^3_{T_3}$	$\partial^5\varphi^3_{T_3}$	$[8-\frac{3}{2}\varepsilon]+\frac{4}{3(N+8)}\varepsilon$	ε^2[99]	N^{-1}[1, 116]	
`Op[Tm[3],6,1]`	$\partial^6\varphi^3_{T_3}$	$\partial^6\varphi^3_{T_3}$	$[9-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^2[99]	N^{-1}[1, 116]	
`Op[Tm[3],6,2]`	$\partial^6\varphi^3_{T_3}$	$\partial^6\varphi^3_{T_3}$	$[9-\frac{3}{2}\varepsilon]+\frac{18}{7(N+8)}\varepsilon$	ε^2[99]	N^{-1}[1, 116]	
`Op[Tm[3],7,1]`	$\partial^7\varphi^3_{T_3}$	$\partial^7\varphi^3_{T_3}$	$[10-\frac{3}{2}\varepsilon]+\frac{3}{2(N+8)}\varepsilon$	ε^2[1, 99]	N^{-1}[1, 116]	

Table 25: Some operators in the rank 4 traceless-symmetric representation. Writing "[1, 99]" means that this particular result was not listed in [99], but was computed using the method of that paper, and likewise for "[1, 116]".

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[Tm[4],0,1]`	$\varphi^4_{T_4}$	$\varphi^4_{T_4}$	$[4-2\varepsilon]+\frac{12}{N+8}\varepsilon$	ε^6[60]	N^{-2}[62]	$\mathtt{3}_4$, $\mathtt{10}_{0,4}$, $\mathtt{11}_{4,0}$
`Op[Tm[4],0,2]`	$\varphi^6_{T_4}$	$\sigma\varphi^4_{T_4}$	$[6-3\varepsilon]+\frac{N+38}{N+8}\varepsilon$	ε^5[5]	N^{-1}[66]	$\mathtt{10}_{1,4}$
`Op[Tm[4],0,3]`	$\Box^2\varphi^4_{T_4}$		$[8-2\varepsilon]+\frac{20}{3(N+8)}\varepsilon$	ε^2[5]		
`Op[Tm[4],0,4]`	$\Box\varphi^6_{T_4}$		$[8-3\varepsilon]+\frac{N+18}{N+8}\varepsilon$	ε^2[5]		
`Op[Tm[4],0,5]`	$\varphi^8_{T_4}$	$\sigma^2\varphi^4_{T_4}$	$[8-4\varepsilon]+\frac{2N+76}{N+8}\varepsilon$	ε^2[5]	N^{-1}[66]	$\mathtt{10}_{2,4}$
`Op[Tm[4],1,1]`	$\partial\varphi^6_{T_4}$		$[7-3\varepsilon]+\frac{N+26}{N+8}\varepsilon$	ε^1		
`Op[Tm[4],2,1]`	$\partial^2\varphi^4_{T_4}$	$\partial^2\varphi^4_{T_4}$	$[6-2\varepsilon]+\frac{26}{3(N+8)}\varepsilon$	ε^2[99]ε^5[5]	N^{-1}[1, 116]	$\mathtt{12}_4$
`Op[Tm[4],3,1]`	$\partial^3\varphi^4_{T_4}$	$\partial^3\varphi^4_{T_4}$	$[7-2\varepsilon]+\frac{6}{N+8}\varepsilon$	ε^2[99]	N^{-1}[1, 116]	
`Op[Tm[4],4,1]`	$\partial^4\varphi^4_{T_4}$	$\partial^4\varphi^4_{T_4}$	$[8-2\varepsilon]+\frac{8}{3(N+8)}\varepsilon$	ε^2[1, 99]	N^{-1}[1, 116]	
`Op[Tm[4],4,2]`	$\partial^4\varphi^4_{T_4}$	$\partial^4\varphi^4_{T_4}$	$[8-2\varepsilon]+\frac{38}{5(N+8)}\varepsilon$	ε^2[1, 99]	N^{-1}[1, 116]	
`Op[Tm[4],5,1]`	$\partial^5\varphi^4_{T_4}$	$\partial^5\varphi^5_{T_4}$	$[9-2\varepsilon]+\frac{6}{N+8}\varepsilon$	ε^2[1, 99]	N^{-1}[1, 116]	
`Op[Tm[4],6,1]`	$\partial^6\varphi^4_{T_4}$	$\partial^6\varphi^4_{T_4}$	$[10-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^2[1, 99]	N^{-1}[1, 116]	
`Op[Tm[4],6,2]`	$\partial^6\varphi^4_{T_4}$	$\partial^6\varphi^4_{T_4}$	$[10-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^2[1, 99]	N^{-1}[1, 116]	
`Op[Tm[4],6,3]`	$\partial^6\varphi^4_{T_4}$	$\partial^6\varphi^4_{T_4}$	$[10-2\varepsilon]+0\varepsilon N^0_{\dots}$	ε^2[1, 99]	N^{-1}[1, 116]	

For the leading scalar T_3 operator `Op[Tm[3],0,1]`, we report ε^3[117]ε^4[110]ε^6[60] and N^{-1}[112]N^{-2}[62]. Numerical values in $d=3$ are $\Delta_{\varphi^3_{T_3}} = 2.1086(3)$ for $N=2$ [76], $\Delta_{\varphi^3_{T_3}} = 1.0384(10)$ for $N=3$ [89], and $\Delta_{\varphi^3_{T_3}} = 1.9768(10)$ for $N=4$ [89].

Table 26: Scalar operators in various O(N) representations.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[Hm[3],0,1]`	$\Box\varphi^5_{H_3}$		$[7-\frac{5}{2}\varepsilon]+\frac{N+13}{N+8}\varepsilon$	ε^1		
`Op[B4,0,1]`	$\Box\varphi^4_{B_4}$	$\Box\varphi^4_{B_4}$	$[6-2\varepsilon]+\frac{6}{N+8}\varepsilon$	ε^5 [5]		
`Op[B4,0,2]`	$\Box^2\varphi^4_{B_4}$		$[8-2\varepsilon]+\frac{8}{3(N+8)}\varepsilon$	ε^2 [5]		
`Op[B4,0,3]`	$\Box\varphi^6_{B_4}$		$[8-3\varepsilon]+\frac{N+24}{N+8}\varepsilon$	ε^2 [5]		
`Op[Hm[4],0,1]`	$\Box\varphi^6_{H_4}$		$[8-3\varepsilon]+\frac{N+22}{N+8}\varepsilon$	ε^2 [5]		
`Op[Tm[5],0,1]`	$\varphi^5_{T_5}$	$\varphi^5_{T_5}$	$[5-\frac{5}{2}\varepsilon]+\frac{20}{N+8}\varepsilon$	ε^5 [1, 60, 110] ε^6 [111]	N^{-2} [62]	$\mathtt{3}_5$, $\mathtt{10}_{0,5}$, $\mathtt{11}_{5,0}$
`Op[Tm[5],0,2]`	$\varphi^7_{T_5}$	$\sigma\varphi^5_{T_7}$	$[7-\frac{7}{2}\varepsilon]+\frac{N+52}{N+8}\varepsilon$	ε^1	N^{-1} [66]	$\mathtt{10}_{1,6}$
`Op[Tm[6],0,1]`	$\varphi^6_{T_6}$	$\varphi^6_{T_6}$	$[6-3\varepsilon]+\frac{30}{N+8}\varepsilon$	ε^5 [1, 60, 110] ε^6 [111]	N^{-2} [62]	$\mathtt{3}_6$, $\mathtt{10}_{0,6}$, $\mathtt{11}_{6,0}$
`Op[Tm[6],0,2]`	$\varphi^8_{T_6}$	$\sigma\varphi^6_{T_6}$	$[8-3\varepsilon]+\frac{N+68}{N+8}\varepsilon$	ε^2 [5]	N^{-1} [66]	$\mathtt{10}_{1,6}$
`Op[YT[{3,2}],0,1]`	$\Box\varphi^5_{Y_{3,2}}$		$[7-\frac{3}{2}\varepsilon]+\frac{12}{N+8}\varepsilon$	ε^1		
`Op[YT[{4,2}],0,1]`	$\Box\varphi^6_{Y_{4,2}}$		$[8-3\varepsilon]+\frac{20}{N+8}\varepsilon$	ε^2 [5]		

Table 27: Spinning operators in various O(N) representations.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\Delta^{(1)}_{4-\varepsilon}$	$\Delta(\varepsilon)$	$\Delta(N)$	Family
`Op[Hm[3],1,1]`	$\partial\varphi^3_{H_3}$	$\partial\varphi^3_{H_3}$	$[4-\frac{3}{2}\varepsilon]+\frac{3}{N+8}\varepsilon$	ε^5[5]	N^{-1}[113]	$11_{2,1}$
`Op[Hm[3],1,2]`	$\partial\varphi^5_{H_3}$	$\partial\sigma\varphi^3_{H_3}$	$[6-\frac{5}{2}\varepsilon]+\frac{N+19}{N+8}\varepsilon$	ε^5[5]		
`Op[Hm[3],2,1]`	$\partial^2\varphi^3_{H_3}$	$\partial^2\varphi^3_{H_3}$	$[5-\frac{3}{2}\varepsilon]+\frac{4}{3(N+8)}\varepsilon$	ε^5[5]		
`Op[Hm[3],2,2]`	$\partial^2\varphi^5_{H_3}$		$[7-\frac{5}{2}\varepsilon]+\frac{3N+35}{3(N+8)}$	ε^1		
`Op[Hm[3],2,3]`	$\partial^2\varphi^5_{H_3}$		$[7-\frac{5}{2}\varepsilon]+\frac{3N+50}{3(N+8)}$	ε^1		
`Op[Hm[3],3,1]`	$\partial^3\varphi^3_{H_3}$	$\partial^3\varphi^3_{H_3}$	$[6-\frac{3}{2}\varepsilon]+\frac{5}{2(N+8)}\varepsilon$	ε^1		
`Op[Hm[3],4,1]`	$\partial^4\varphi^3_{H_3}$	$\partial^4\varphi^3_{H_3}$	$[7-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[Hm[3],4,2]`	$\partial^4\varphi^3_{H_3}$	$\partial^4\varphi^3_{H_3}$	$[7-\frac{3}{2}\varepsilon]+\frac{8}{5(N+8)}\varepsilon$	ε^1		
`Op[Hm[3],5,1]`	$\partial^5\varphi^3_{H_3}$	$\partial^5\varphi^3_{H_3}$	$[8-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[Hm[3],5,2]`	$\partial^5\varphi^3_{H_3}$	$\partial^5\varphi^3_{H_3}$	$[8-\frac{3}{2}\varepsilon]+\frac{7}{3(N+8)}\varepsilon$	ε^1		
`Op[Hm[3],6,1]`	$\partial^6\varphi^3_{H_3}$	$\partial^6\varphi^3_{H_3}$	$[9-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[Hm[3],6,2]`	$\partial^6\varphi^3_{H_3}$	$\partial^6\varphi^3_{H_3}$	$[9-\frac{3}{2}\varepsilon]+\frac{12}{7(N+8)}\varepsilon$	ε^1		
`Op[A3,3,1]`	$\partial^3\varphi^3_{A_3}$	$\partial^3\varphi^3_{A_3}$	$[6-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[A3,5,1]`	$\partial^5\varphi^3_{A_3}$	$\partial^5\varphi^3_{A_3}$	$[8-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[A3,6,1]`	$\partial^6\varphi^3_{A_3}$	$\partial^6\varphi^3_{A_3}$	$[9-\frac{3}{2}\varepsilon]+0\varepsilon$	ε^1		
`Op[Hm[4],1,1]`	$\partial\varphi^4_{H_4}$	$\partial\varphi^4_{H_4}$	$[5-2\varepsilon]+\frac{8}{N+8}\varepsilon$	ε^5[5]	N^{-1}[113]	$11_{3,1}$
`Op[Hm[4],1,2]`	$\partial\Box\varphi^4_{T_4}$		$[7-2\varepsilon]+\frac{16}{3(N+8)}\varepsilon$	ε^1		
`Op[Hm[4],1,3]`	$\partial\varphi^6_{T_4}$		$[7-3\varepsilon]+\frac{N+30}{N+8}\varepsilon$	ε^1		
`Op[Hm[4],2,1]`	$\partial^2\varphi^4_{H_4}$	$\partial^2\varphi^4_{H_4}$	$[6-2\varepsilon]+\frac{6}{N+8}\varepsilon$	ε^5[5]		
`Op[Hm[4],3,1]`	$\partial^3\varphi^4_{H_4}$	$\partial^3\varphi^4_{H_4}$	$[7-2\varepsilon]+\frac{10}{3(N+8)}\varepsilon$	ε^1		
`Op[Hm[4],3,2]`	$\partial^3\varphi^4_{H_4}$	$\partial^3\varphi^4_{H_4}$	$[7-2\varepsilon]+\frac{7}{N+8}\varepsilon$	ε^1		
`Op[B4,2,1]`	$\partial^2\varphi^4_{B_4}$	$\partial^2\varphi^4_{B_4}$	$[6-2\varepsilon]+\frac{14}{3(N+8)}\varepsilon$	ε^5[5]	N^{-1}[113]	$11_{2,2}$
`Op[Tm[5],2,1]`	$\partial^2\varphi^5_{T_5}$	$\partial^2\varphi^5_{T_5}$	$[7-\frac{5}{2}\varepsilon]+\frac{16}{N+8}\varepsilon$	ε^2[2, 99]	N^{-1}[2, 116]	12_5
`Op[Hm[5],1,1]`	$\partial\varphi^5_{H_5}$	$\partial\varphi^5_{H_5}$	$[6-\frac{5}{2}\varepsilon]+\frac{15}{N+8}\varepsilon$	ε^5[5]	N^{-1}[113]	$11_{4,1}$
`Op[Hm[5],2,1]`	$\partial^2\varphi^5_{H_5}$	$\partial^2\varphi^5_{H_5}$	$[7-\frac{5}{2}\varepsilon]+\frac{38}{3(N+8)}\varepsilon$	ε^1		
`Op[Hm[6],1,1]`	$\partial\varphi^6_{H_6}$	$\partial\varphi^6_{H_6}$	$[7-3\varepsilon]+\frac{24}{N+8}\varepsilon$	ε^1	N^{-1}[113]	$11_{5,1}$
`Op[Tm[6],2,1]`	$\partial^2\varphi^6_{T_5}$	$\partial^2\varphi^6_{T_6}$	$[8-3\varepsilon]+\frac{76}{3(N+8)}\varepsilon$	ε^2[2, 99]	N^{-1}[2, 116]	12_6
`Op[YT[{3,2}],2,1]`	$\partial^2\varphi^5_{Y_{3,2}}$	$\partial^2\varphi^5_{Y_{3,2}}$	$[7-\frac{5}{2}\varepsilon]+\frac{32}{3(N+8)}\varepsilon$	ε^1		
`Op[YT[{2,1,1}],1,1]`	$\partial\Box\varphi^4_{Y_{2,1,1}}$		$[7-2\varepsilon]+\frac{8}{3(N+8)}\varepsilon$	ε^1		
`Op[YT[{2,1,1}],3,1]`	$\partial^3\varphi^4_{Y_{2,1,1}}$		$[7-2\varepsilon]+\frac{11}{3(N+8)}\varepsilon$	ε^1		

3.5 Operators in non-traceless-symmetric Lorentz representations

Here we will just mention a few operators in non-traceless-symmetric Lorentz representations, in order to give a complete presentation of the spectrum up to $\Delta^{\text{4d}} \leqslant 6$.

Table 28: $\mathrm{O}(N)$ singlet and Lorentz non-traceless-symmetric operators up to 4d dimension 8. The 4d irrep notation is the one used in [95].

$\mathcal{O}$	Lorentz YT	4d irrep	Field content	$\Delta^{(1)}_{4-\varepsilon}$
`Op[S,{2,1},1]`	$\ydiagram{2,1}$	$(\frac{3}{2},\frac{1}{2})+(\frac{1}{2},\frac{3}{2})$	$\partial^3\varphi^4_S$	$[7-2\varepsilon]+\frac{N+2}{N+8}\varepsilon$
`Op[S,{3,1},1]`	$\ydiagram{3,1}$	$(2,1)+(1,2)$	$\partial^4\varphi^4_S$	$[8-2\varepsilon]+\frac{N+2}{N+8}\varepsilon$
`Op[S,{2,2},1]`	$\ydiagram{2,2}$	$(2,0)+(0,2)$	$\partial^4\varphi^4_S$	$[8-2\varepsilon]+\frac{3N+20-\sqrt{9N^2+64N+288}}{6(N+8)}\varepsilon$
`Op[S,{2,2},2]`	$\ydiagram{2,2}$	$(2,0)+(0,2)$	$\partial^4\varphi^4_S$	$[8-2\varepsilon]+\frac{3N+20+\sqrt{9N^2+64N+288}}{6(N+8)}\varepsilon$

Dimension $\Delta = 5$ One operator in $\mathrm{O}(N)$ irrep A_3, Lorentz irrep $\ydiagram{1,1}$: $\varphi^{[i}\partial_{[\mu}\varphi^j\partial_{\nu]}\varphi^{k]}$, dimension $\Delta = [5-\frac{3}{2}\varepsilon]+\ldots+O(\varepsilon^6)$, where we have ε^5[5]. Implemented as `Op[A3,{1,1},1]`. Note that this anomalous dimensions is negative at $O(\varepsilon^2)$ and $O(\varepsilon^3)$.

Dimension $\Delta = 6$ Two operators constructed out of three fields and three gradients: $\mathrm{O}(N)$ irreps V and H_3, both in Lorentz irrep $\ydiagram{2,1}$, and both with dimension $\Delta = [6-\frac{3}{2}\varepsilon]+O(\varepsilon^2)$. Moreover, two operators constructed out of four fields and two gradients, both in Lorentz irrep $\ydiagram{1,1}$: $\mathrm{O}(N)$ irrep A, dimension $\Delta = [6-2\varepsilon]+\frac{N+2}{N+8}\varepsilon+\ldots+O(\varepsilon^6)$; $\mathrm{O}(N)$ irrep $Y_{2,1,1}$, dimension $\Delta = [6-2\varepsilon]+\frac{4}{N+8}\varepsilon+\ldots+O(\varepsilon^6)$. For the latter two we have ε^5[5]. These operators are implemented as

$$\texttt{Op[V,\{2,1\},1]},\ \texttt{Op[Hm[3],\{2,1\},1]},\ \texttt{Op[A,\{1,1\},1]},\ \texttt{Op[YT[\{2,1,1\}],\{1,1\},1]}. \tag{3.39}$$

In table 28 we give the leading $\mathrm{O}(N)$ singlet operators in non-traceless-symmetric $\mathrm{O}(N)$ representations, up to $\Delta^{\text{4d}} \leqslant 8$.

Operators in the Lorentz representation labelled by the Young tableau $\ydiagram{1,1}$ are related to parity-odd scalars in the three-dimensional theory. In the free three-dimensional $\mathrm{O}(N)$ symmetric CFT, one can for generic $N > 1$ construct a parity odd singlet scalar out of four fields and five gradients, giving (in the free theory) dimension 7 [118], equation 3.1. It has the form $\epsilon^{\mu\nu\rho}\partial_\mu\partial_\nu\partial_\rho\Box\varphi^4_S$. The corresponding operator in the $4-\varepsilon$ expansion has dimension $[9-2\varepsilon]+0\varepsilon+O(\varepsilon^2)$.

3.6 OPE coefficients in the $\varphi \times \varphi$ OPE

Table 29: OPE coefficients in the $\varphi \times \varphi$ OPE for operators for general N.

$\mathcal{O}$	$\mathcal{O}\vert\varepsilon$	$\mathcal{O}\vert N$	$\lambda^2_{\varphi\varphi\mathcal{O}}(\varepsilon)$	$\lambda^2_{\varphi\varphi\mathcal{O}}(\varepsilon)$	$\lambda^2_{\varphi\varphi\mathcal{O}}(N)$	$\lambda^2_{\varphi\varphi\mathcal{O}}(N)$
$\varphi, \varphi,$ `Op[S,0,0]`	$\mathbb{1}$	$\mathbb{1}$	1	exact	1	exact
$\varphi, \varphi,$ `Op[S,0,1]`	φ_S^2	σ	$O(1)$	ε^3[64]	$O(N^{-1})$	N^{-2}[66]
$\varphi, \varphi,$ `Op[S,0,2]`	φ_S^4	σ^2	$O(\varepsilon^2)$	ε^2[65]	$O(N^{-2})$	N^{-2}[119]
$\varphi, \varphi,$ `Op[S,0,4]`	φ_S^6	σ^3	$O(\varepsilon^4)$	ε^4[120]		
$\varphi, \varphi,$ `Op[S,2,1]`	$T^{\mu\nu}$	$T^{\mu\nu}$	$O(1)$	ε^4[65]	$O(N^{-1})$	N^{-2}[63]
$\varphi, \varphi,$ `Op[S,2,2]`	$\partial^2\varphi_S^4$		$O(\varepsilon^2)$	ε^2[1, 65]		
$\varphi, \varphi,$ `Op[S,2,3]`	$\partial^2\varphi_S^4$	$[\sigma,\sigma]_{0,2}$	$O(\varepsilon^2)$	ε^2[1, 65]	$O(N^{-2})$	N^{-2}[119]
$\varphi, \varphi,$ `Op[S,4,1]`	$\mathcal{J}_{S,4}$	$\mathcal{J}_{S,4}$	$O(1)$	ε^4[65]	$O(N^{-1})$	N^{-2}[119]
$\varphi, \varphi,$ `Op[T,0,1]`	φ_T^2	φ_T^2	$O(1)$	ε^3[64]	$O(1)$	N^{-1}[64, 121]
$\varphi, \varphi,$ `Op[T,0,2]`	φ_T^4		$O(\varepsilon^2)$	ε^2[65]	$O(N^{-1})$	N^{-1}[1, 121]
$\varphi, \varphi,$ `Op[T,2,1]`	$\mathcal{J}_{T,2}$	$\mathcal{J}_{T,2}$	$O(1)$	ε^4[65]	$O(1)$	N^{-1}[64, 121]
$\varphi, \varphi,$ `Op[T,4,1]`	$\mathcal{J}_{T,4}$	$\mathcal{J}_{T,4}$	$O(1)$	ε^4[65]	$O(1)$	N^{-1}[64, 121]
$\varphi, \varphi,$ `Op[A,1,1]`	J^μ	J^μ	$O(1)$	ε^4[65]	$O(1)$	N^{-1}[69]N^{-2}(num)[119]
$\varphi, \varphi,$ `Op[A,1,2]`	$\partial\varphi_A^4$		$O(\varepsilon^2)$	ε^2[65]	$O(N^{-1})$	N^{-1}[1, 121]
$\varphi, \varphi,$ `Op[A,3,1]`	$\mathcal{J}_{A,3}$	$\mathcal{J}_{A,3}$	$O(1)$	ε^4[65]	$O(1)$	N^{-1}[64, 121]

Table 30: Other OPE coefficients implemented for general N.

$\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$	$\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3\|_\varepsilon$	$\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3\|_N$	$\lambda^2_{\mathcal{O}_1\mathcal{O}_2\mathcal{O}_3}\|_\varepsilon$	$\lambda^2_{\mathcal{O}_1\mathcal{O}_2\mathcal{O}_3}\|_N$
`Op[S,0,1],Op[S,0,1],Op[S,0,1]`	$\varphi_S^2, \varphi_S^2, \varphi_S^2$	σ, σ, σ	ε^1[106]	N^{-2}[122]
`Op[S,0,1],Op[S,0,1],Op[S,0,2]`	$\varphi_S^2, \varphi_S^2, \varphi_S^4$	σ, σ, σ^2	ε^1[7]	N^{-1}(3d)[123]
`Op[T,0,1],Op[T,0,1],Op[S,0,1]`	$\varphi_T^2, \varphi_T^2, \varphi_S^2$	$\varphi_T^2, \varphi_T^2, \sigma$	ε^1[7]	
`Op[T,0,1],Op[T,0,1],Op[S,0,2]`	$\varphi_T^2, \varphi_T^2, \varphi_S^4$	$\varphi_T^2, \varphi_T^2, \sigma^2$	ε^1[7]	
`Op[T,0,1],Op[T,0,1],Op[T,0,1]`	$\varphi_T^2, \varphi_T^2, \varphi_T^2$	$\varphi_T^2, \varphi_T^2, \varphi_T^2$	ε^1[7]	
`Op[T,0,1],Op[T,0,1],Op[T,0,2]`	$\varphi_T^2, \varphi_T^2, \varphi_T^4$	$\varphi_T^2, \varphi_T^2, \sigma\varphi_T^2$	ε^1[7]	
`Op[T,0,1],Op[T,0,1],Op[Tm[4],0,1]`	$\varphi_T^2, \varphi_T^2, \varphi_{T_4}^4$	$\varphi_T^2, \varphi_T^2, \varphi_{T_4}^4$	ε^1[7]	
`Op[T,0,1],Op[T,0,1],Op[B4,0,1]`	$\varphi_T^2, \varphi_T^2, \Box\varphi_{B_4}^4$	$\varphi_T^2, \varphi_T^2, \Box\varphi_{B_4}^4$	ε^1[7]	

A Some extension on representation theory

We would like to give some more details on some additional $O(N)$ representations. We reproduce table 7 with some additional irreps displayed

Table 7: Dimensions of the irreducible representations of O(N) symmetry with rank up to 4. For $N = 2$ and $N = 3$ we also give additional symbols for the irreps, corresponding to those used in [124] and [77]. For $N = 3$, the parity corresponds to the parity of the number of boxes in the Young tableau.

Irrep			$N = 2$		$N = 3$		$N = 4$	$N \geqslant 5$
S	$\bullet$	`S`	$q^p = 0^+$	1	$\mathbf{q}^p = 0^+$	1	1	1
V	$\ydiagram{1}$	`V`	$q = 1$	2	$\mathbf{q}^p = 1^-$	3	4	N
A	$\ydiagram{1,1}$	`A`	$q^p = 0^-$	1	$\mathbf{q}^p = 1^+$	3	$3 + \overline{3}$	$N(N-1)/2$
T	$\ydiagram{2}$	`T`	$q = 2$	2	$\mathbf{q}^p = 2^+$	5	9	$(N-1)(N+2)/2$
A_3	$\ydiagram{1,1,1}$	`A3`			$\mathbf{q}^p = 0^-$	1	4^-	$N(N-1)(N-2)/6$
T_3	$\ydiagram{3}$	`Tm[3]`	$q = 3$	2	$\mathbf{q}^p = 3^-$	7	16	$N(N+4)(N-1)/6$
H_3	$\ydiagram{2,1}$	`Hm[3]`			$\mathbf{q}^p = 2^-$	5	$8 + \overline{8}$	$N(N+2)(N-2)/3$
A_4	$\ydiagram{1,1,1,1}$	—					1^-	$N(N-1)(N-2)(N-3)/24$
T_4	$\ydiagram{4}$	`Tm[4]`	$q = 4$	2	$\mathbf{q}^p = 4^+$	9	25	$N(N+1)(N+6)(N-1)/24$
H_4	$\ydiagram{3,1}$	`Hm[4]`			$\mathbf{q}^p = 3^+$	7	$15 + \overline{15}$	$(N+1)(N+4)(N-1)(N-2)/8$
B_4	$\ydiagram{2,2}$	`B4`					$5 + \overline{5}$	$N(N+1)(N+2)(N-3)/12$
$Y_{2,1,1}$	$\ydiagram{2,1,1}$	`YT[{2,1,1}]`					9^-	$N(N+2)(N-1)(N-3)/8$
T_5	$\ydiagram{5}$	`Tm[5]`	$q = 5$	2	$\mathbf{q}^p = 5^-$	11	36	$N(N+1)(N+2)(N+8)(N-1)/120$
T_6	$\ydiagram{6}$	`Tm[6]`	$q = 6$	2	$\mathbf{q}^p = 6^+$	13	48	$N(N+1)(N+2)(N+3)(N+10)(N-1)/720$
H_5	$\ydiagram{4,1}$	`Hm[5]`			$\mathbf{q}^p = 4^-$	9	49	$N(N+2)(N+6)(N-1)(N-2)/30$
$Y_{3,2}$	$\ydiagram{3,2}$	`YT[{3,2}]`					$12 + \overline{12}$	$N(N+2)(N+4)(N-1)(N-3)/24$
$Y_{4,2}$	$\ydiagram{4,2}$	`YT[{4,2}]`					$21 + \overline{21}$	$N^2(N+3)(N+6)(N-1)(N-3)/80$

Note that the dimension of the T_m irrep is $\frac{(2m+n-2)\Gamma(m+n-2)}{\Gamma(m+1)\Gamma(n-1)}$.

B Extended character decomposition

For an $O(N)$ irrep labelled by Young tableau Y_λ, $\lambda = (\lambda_1, \ldots, \lambda_r)$, the characters can be computed using equation (A.7) of [125] (based on [126]. The rest of the computation is described in appendix B of [1]. Other relevant references are appendix A of [3], equation (2.57) of [127] (for the extraction of singlets).

We reproduce here the scalar spectrum:

$$\begin{aligned}
Z_0 = {} & S + \Delta V\phi + \Delta^2(S+T)\phi^2 + \Delta^3(V_{\text{EOM}} + T_3)\phi^3 + \Delta^4(S+T+T_4)\phi^4 \\
& + \Delta^5(V + T_3 + T_5)\phi^5 \\
& + \Delta^6\big[(S+T+B_4)\phi^4 + (S+T+T_4+T_6)\phi^6\big] \\
& + \Delta^7\big[(V + H_3 + T_3 + Y_{3,2})\phi^5 + (V + T_3 + T_5 + T_7)\phi^7\big] \\
& + \Delta^8\big[(2S + 2T + T_4 + B_4)\phi^4 + (S + 2T + H_4 + T_4 + B_4 + Y_{4,2})\phi^6 \\
& \qquad + (S + T + T_4 + T_6 + T_8)\phi^8\big] \\
& + \Delta^9\big[(4V + 3T_3 + T_5 + 3H_3 + H_5 + Y_{3,2} + Y_{2,2,1})\phi^5 \\
& \qquad + (V + 2T_3 + T_5 + H_3 + H_5 + Y_{3,2} + Y_{5,2})\phi^7 + (V + T_3 + T_5 + T_7 + T_9)\phi^9\big] \\
& + \Delta^{10}\big[(2S + 2T + T_4 + A_4 + B_4)\phi^4 \\
& \qquad + (4S + 8T + 4T_4 + T_6 + 2A + 4B_4 + 4H_4 + H_6 + 2Y_{4,2} + Y_{2,1,1} + Y_{2,2,2} + Y_{3,2,1})\phi^6 \\
& \qquad + (S + 2T + 2T_4 + T_6 + H_4 + H_6 + B_4 + Y_{4,2} + Y_{6,2})\phi^8 \\
& \qquad + (S + T + T_4 + T_6 + T_8 + T_{10})\phi^{10}\big] \\
& + O(\Delta^{11})
\end{aligned} \tag{B.1}$$

Finite-N effects can be checked by replacing the general characters by those for an $O(N)$ group with finite rank. In this way, one can reproduce some identifies from [125], and generate new. We have not implemented this systematically. Examples are

$$(2,2)|_{N=2} = -(2) \tag{B.2}$$

$$(4,2)|_{N=2} = -(4) \tag{B.3}$$

where the latter was confirmed in private communication with W. Cao and T. Melia.

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